COURSE SLIDES · WSQ

Agentic AI for Business Process Automation

WSQ Course Code: TGS-2024045801

TSC: Artificial Intelligence Application · AER-TEM-3026-1.1

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1

AM, PM and Assessment

It is mandatory to take all three digital attendances for WSQ-funded courses.

2

The QR code

The trainer or administrator displays the attendance QR code from the SSG portal.

3

Scan and submit

Scan it with your mobile phone camera and submit your attendance.

4

75% minimum

At least 75% attendance is required to be eligible for assessment and funding.

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE
Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS
PhD — specialises in AI, automation and software engineering.

DELIVERS
WSQ courses on agentic AI, workflow automation (n8n) and app development.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

1

Who you are

Your name, your organisation and your role.

2

Your starting point

Your experience with automation or AI tools — including none at all.

3

Your pain point

One business process in your work you would most like to automate.

4

Your goal

What you want to be able to build by the end of this course.

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HOUSEKEEPING

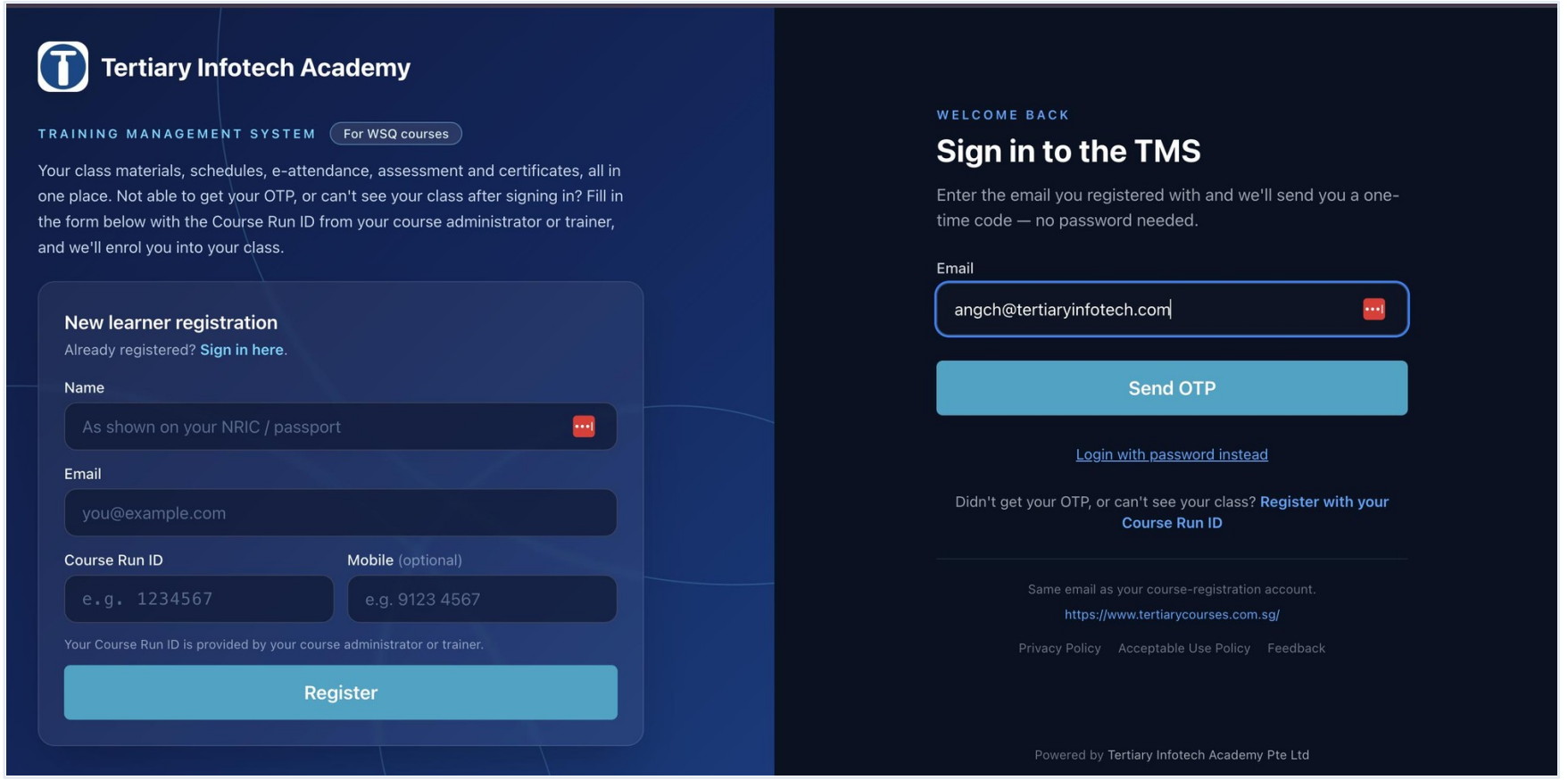
Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

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COURSE PORTAL

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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COURSE PORTAL · STEPS

Download Course Material

1. Go to lms-tms.tertiaryinfotech.com
2. Enter your registered email and click Send OTP
3. Key in the one-time password from your inbox
4. Open this course and download the slides + Learner Guide
5. Keep them open — the assessment is open book

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GET THE FLOWS

Get the Course Workflows

1

Every lab ships a workflow

A ready-made n8n workflow (.json) sits in each lab's folder under labs/.

2

Import it

In n8n: Workflows → Add workflow → the '...' menu → Import from File → pick the lab .json.

3

Re-select your credentials

After importing, re-select YOUR OWN credentials on every node showing a warning.

4

Nothing of yours is shared

Imported workflows reference credential NAMES, not secrets — no key travels in the file.

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WSQ ALIGNMENT

Skills Framework (TSC)

1

TSC Title

Artificial Intelligence Application

2

TSC Code

AER-TEM-3026-1.1

3

Mapped to the national framework

This WSQ course is mapped to the SkillsFuture Singapore Skills Framework.

4

Fully assessed

The assessment covers every Ability (A) and Knowledge (K) statement listed next.

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SKILLS FRAMEWORK

TSC Abilities and Knowledge

ABILITIES (assessed by the PP)

- A1 Implement AI applications in collaboration with technology service providers
- A2 Deploy AI workflows according to plan
- A3 Identify and report any issues with the AI applications and data collected

KNOWLEDGE (assessed by the WA)

- K1 Background and basics of AI
- K2 Fundamental concepts and methods of statistics and programming for data science
- K3 Mathematics and computing theories
- K4 Machine learning techniques and applications
- K5 Principles of data management
- K6 Software development methodologies
- K7 Types of AI applications
- K8 AI implementation procedures

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SCHEDULE

Lesson Plan — 2 Days, 8 hours/day

DAY 1 — n8n, AI Agents and Agentic Chatbots

- Topic 1: Why agentic AI, and why workflow first
- n8n fundamentals → Lab 1 — trigger, action and QR flyer
- AI agent building blocks and webhooks
- Lab 2 — Webhook and AI Agent
- Timing: 9:30 am – 6:30 pm

DAY 2 — Human in the Loop, RAG and Assessment

- Topic 2: Agentic chatbot and human in the loop
- Lab 3 — Chatbot webhook
- Lab 4 — Human in the loop
- Topic 3: RAG
- Lab 5 — RAG customer care chatbot
- Final Assessment (WA + PP)
- Timing: 9:30 am – 6:30 pm instruction, assessment to 8:45 pm

Each day is 8 instructional hours: a 1-hour lunch is excluded, and the two 15-minute tea breaks are counted within the instructional time

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 · n8n and AI agents

Explain what n8n and AI agents are, and build workflows using triggers and actions.

2

LO2 · Webhook-triggered agents

Implement AI agent applications triggered by webhooks and integrated with business systems.

3

LO3 · Agentic chatbots

Design agentic chatbot workflows that operate within business and regulatory constraints.

4

LO4 · Human in the loop

Apply human approval so that sensitive actions are authorised by a named person.

5

LO5 · RAG

Deploy Retrieval-Augmented Generation to ground an agent in your organisation's documents.

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COURSE JOURNEY

Five Connected Labs

Each lab keeps the agent and changes one thing — the front door, the tools, the oversight, the knowledge.

LAB 1
form → email

▶

LAB 2
webhook + agent + tools

▶

LAB 3
chatbot + memory

▶

LAB 4
human approval gate

▶

LAB 5
RAG over documents

1 The front door changes

Form Trigger in Lab 1 becomes a Webhook in Labs 2–5 — the workflow behind it barely moves.

2 The agent gains abilities

Lab 2 adds tools, Lab 3 adds memory, Lab 5 adds retrieval over your own documents.

3 The oversight grows

Lab 4 stops the workflow dead until a licensed person approves what the agent drafted.

By Lab 5 you have built the whole pattern: a front door, an agent, its tools, its guardrails and its knowledge.

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WSQ FUNDING

Criteria for Funding

1

75% attendance

A minimum attendance rate of 75%, based on the SSG Digital Attendance record.

2

Assessed Competent

Complete the assessment and be assessed as 'Competent'.

3

Both are required

Both criteria must be met for the course fee funding to be granted.

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ASSESSMENT BRIEFING

Briefing for Assessment

1. Place phones and other materials under the table or on the floor.
2. No photos or recording of assessment scripts.
3. No discussion during the assessment.
4. Use a black or blue pen for hard-copy assessments.
5. No liquid paper or correction tape.
6. Scripts are collected when time is up.

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FINAL ASSESSMENT

Assessment

1. **Written Assessment (SAQ)**
   1 hour. Short-answer questions covering the course knowledge.

2. **Practical Performance (PP)**
   1 hour. Hands-on n8n build tasks drawn from the five labs.

3. **Open book**
   Slides, Learner Guide and approved materials only.

4. **75% attendance**
   A minimum of 75% attendance is required to be eligible for assessment and funding.

5. **Appeals**
   An appeal process is available if required.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

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01

TOPIC 01

Overview of n8n and AI Agents

Why agentic AI · Workflow first · Triggers & actions · LLM agents · Webhooks

Applied in Lab 1, Lab 2

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A Brief History of AI, 2023–2026

Each year moved the skill that matters — from the words you type to the system you build.

Prompt Engineering
2023

▶

Context Engineering
2024

▶

Harness Engineering
2025

▶

AI Agents
2026

Wording the prompt

ChatGPT goes mainstream. Results depend on how well you word the prompt.

Curating the context

Models call tools and MCP launches. Results depend on the documents, data and tools you supply.

Building the harness

Claude Code and Codex: systems that plan, act and check their own work in a loop.

Agents at work

Always-on agents work 24/7 — OpenClaw, Hermes Agent and agent teams take on whole jobs.

From wording one prompt → curating context → engineering the harness → AI agents that work around the clock.

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Generative AI vs Agentic AI

The difference is not how clever the model is — it is whether it can act.

GENERATIVE AI — answers

- You ask; it writes one answer.
- No access to your systems, and no actions.
- Quality depends on what you paste into the prompt.
- Stops the moment the answer is produced.
- Example: draft me an email to this customer.

AGENTIC AI — acts

- Works toward a goal, not just one reply.
- Calls tools: looks up data, calls APIs, writes records.
- Holds a conversation with memory.
- Continues until the goal is met.
- Example: read the enquiry, check the CRM, reply and log it.

Generative AI produces content. Agentic AI decides and does — which is exactly why it needs governance.

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Agentic AI vs AI Agents

AI agents are a special type of agentic AI system — the kind that never switches off.

Agentic AI — the broad category

- Any AI system that pursues a goal over many steps: plans, acts through tools, checks the result
- Usually started by a person or a trigger, and stops when the task is done
- Built as a workflow with AI steps, or around an LLM loop
- Includes the orchestration, guardrails and human approvals
- Examples: a Claude Code or Codex session; an n8n workflow with AI steps

AI Agents — always-on agentic AI

- A special type of agentic AI system that exists 24/7
- Usually installed in the cloud, so it serves all the time — no laptop needs to be open
- Waits for events — messages, emails, webhooks, schedules — and acts on them around the clock
- Keeps its memory across conversations and days
- Examples: OpenClaw, Hermes Agent, an n8n agent on a cloud instance behind a webhook

Every AI agent is agentic AI, but not every agentic AI is an AI agent — an AI agent is always on, serving 24/7 from the cloud.

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Workflow First vs Agent First

Workflow first evolves rule-based (RPA) automation: the workflow owns the path, and its steps can be AI or AI agents, as in n8n.

	Workflow First	Agent First
Who decides the path	The workflow — steps may be AI nodes or AI Agent nodes	The agent decides the steps and tools at run time
Behaviour	Deterministic — same input, same path, every time	Probabilistic — the path can change between runs
Efficiency	More effort to design every branch up front	Efficient — one instruction replaces many hand-built branches
Reliability & testing	Easy to test, debug and certify step by step	Harder to test; edge cases surface in production
Cost & speed	An LLM call only where needed — cheap and fast	An LLM call at every step — more tokens, more latency
Best for	Repeatable, regulated, high-volume processes	Open-ended tasks with unpredictable inputs

Source: n8n, In The Loop 2026 — opening keynote and product direction (youtube.com/watch?v=KQRe5UurLQ0)

Workflow first is more deterministic; agent first is more efficient. Companies make as much deterministic as they can.

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TOPIC 1 · THE MODERN VIEW

The Trend: AI-Assisted Workflow Creation

AI-powered coding has flipped the trade-off — the agent's efficiency goes into BUILDING the workflow, not running it.

Describe
the process in plain English

▶

n8n Assistant
or Claude Code / Codex drafts it

▶

You review
test and approve it

▶

Workflow runs
deterministic and logged

▶

AI and agent nodes
where judgement is needed

1 Companies want determinism

Reliability, compliance and audit push businesses to make as much of a process deterministic as possible.

2 AI coding made it cheap

n8n Assistant, Claude Code and Codex are AI agents themselves — they make designing, building and changing workflows fast and cheap.

3 The best of both

An AI agent BUILDS the workflow; a deterministic workflow RUNS the process, with AI and AI Agent nodes where judgement is needed.

The trend: AI-assisted workflow creation — the AI builds the workflow, and the workflow runs the business process.

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TOPIC 1 · COMPLEXITY VS CUSTOMISATION

Where AI Agent Platforms Sit

Three tiers — and AI now helps build at every one of them. This course teaches the middle one.

1 No-code

HOW YOU BUILD

Drag and drop in a fully visual builder.

CUSTOMISATION

Lowest. Fastest route to a working agent, least control over the internals.

2 Low-code · n8n

HOW YOU BUILD

AI-assisted: n8n Assistant, Claude Code or Codex drafts the workflow on a visual canvas.

CODE WHEN NEEDED

Add JavaScript or Python in a Code node where a built-in node is not enough.

CUSTOMISATION

High. 400+ integrations, self-host or cloud. This course's platform.

3 Full-code

HOW YOU BUILD

AI generates the code — Claude Code or Codex write it, instead of coding manually.

CUSTOMISATION

Highest — the most customisable tier. SDKs and frameworks such as LangChain, with real engineering effort to test and run.

Less control, less effort · more customisation, more engineering effort →

AI drafts the workflow in low-code and writes the code in full-code. n8n: analysts build it, engineers extend it.

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Business Value of Workflow-First Agentic Automation

The modern view: what a business gains when AI builds the workflow and the workflow runs the process.

1 Reliable and predictable

The same input follows the same path every time, so the process can be tested and trusted like software.

2 Secure by design

Each step holds only the data and credentials it needs; risky actions wait at an approval gate.

3 Auditable end to end

Every run is logged node by node — you can explain, replay and correct any decision.

4 Lower running cost

Deterministic steps cost nothing in tokens; the LLM is called only where judgement is needed.

5 Faster to build and change

Claude Code and Codex draft the workflow in minutes; your team reviews, tests and owns it.

6 Scale around the clock

Triggers respond in real time, 24/7 — ten times the volume costs compute, not ten more hires.

AI where judgement is needed, workflow everywhere else — reliability you can audit, at a cost you can predict.

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TOPIC 1 · USE CASES

Where Agentic Automation Is Used

The same pattern — trigger, agent, tools, oversight — recurs across every function.

1 Customer service

Answer enquiries from your own documents, capture leads, and escalate what needs a person.

2 Finance & accounting

Invoice processing, bank reconciliation, accounts-payable routing and exception handling.

3 Risk & compliance

Screen applications, flag anomalies, and route sensitive cases for human review.

4 HR & people ops

Onboarding, leave applications, policy questions and approval chains.

5 Sales & marketing

Lead qualification, follow-up sequencing, CRM enrichment and content drafting.

6 Operations

Order processing, status notifications, document generation and reporting.

Every one of these is a front door, an agent, some tools and a decision about where the human belongs.

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TOPIC 1 · AI SECURITY

5 Principles of AI Security

Agents act on your behalf. These five controls keep what they can see and do safe, reviewable and accountable.

1

Data Control

WHAT AI CAN SEE

Classify your data — public, internal, confidential — and decide exactly which data the AI may access.

2

Permission Control

WHAT AI CAN DO

Grant only the rights needed — read, write, edit, delete. Ask before anything destructive.

3

Human in the Loop

WHO APPROVES

A person reviews and approves high-impact actions — payments, deletions, customer emails.

4

Traceability

WHAT HAPPENED

Log every prompt, tool call and change so actions can be audited, explained and rolled back.

5

Right-Sized Models

NOT THE MOST POWERFUL

The smallest model that does the job carries less autonomy and less risk than the most powerful one.

In n8n: scoped credentials per node, approval gates (Send and Wait), the Executions log, and a small model per AI step.

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TOPIC 1 · AI SECURITY

Workflow First vs Agent First — AI Security

A fixed path limits the blast radius: a manipulated AI step cannot reach tools that are not wired to it.

Principle	Workflow First	Agent First
Data control	Each node sees only the fields wired into it	The agent can reach any data its tools expose
Permission control	Each credential is scoped to one node and one action	The agent holds every tool's rights at once
Human in the loop	The approval gate sits at a fixed, designed step	Depends on the agent choosing to ask
Traceability	Every run is logged node by node in Executions	Reasoning varies run to run and is harder to explain
Right-sized models	A small model for each narrow AI step	One capable model with broad autonomy

The path is fixed, so every control is designed in rather than hoped for — that is why regulated processes stay workflow first.

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What Is n8n?

A workflow-first agentic AI platform: the workflow owns the path, and AI and AI Agent nodes sit inside it.

1 Build visually

Drag nodes onto a canvas and wire them together. 400+ app integrations, plus HTTP and Code nodes when no integration exists.

2 Run anywhere

Free cloud trial at n8n.io, or self-host with Docker. Webhooks, schedules and chat triggers work the same in both.

3 AI-native

AI Agent and LangChain nodes are built in — attach any LLM (Gemini, OpenAI, Claude), memory, tools and vector stores.

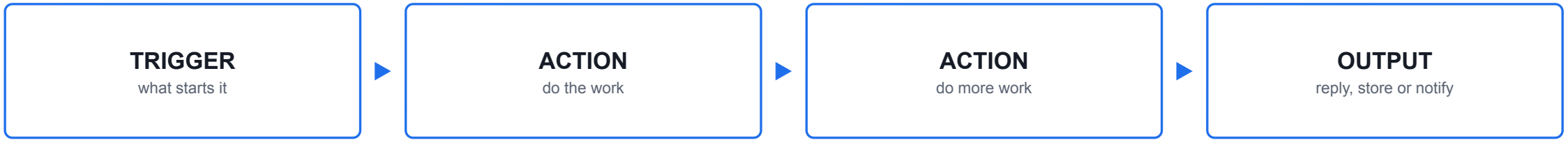
n8n is workflow-first agentic AI — deterministic workflows with AI and agents built in, on one canvas.

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TOPIC 1 · WHAT IS n8n

Every Workflow Is the Same Shape

Trigger first, then actions in order, with data flowing left to right. Only the nodes change.



1 One trigger

A workflow has exactly one entry point. Change the trigger and you have a different workflow — the actions may not change at all.

2 Actions are ordered

Each action runs after the one before it, and can read the output of every node that ran before it.

3 Data flows as JSON

Every node passes a list of JSON items to the next — the same structure in every lab of this course.

Trigger, actions, output. Learn this once in Lab 1 and every later lab is a variation.

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TOPIC 1 · TRAINING ACCOUNTS

n8n Training Accounts

Each learner signs in to n8n with ONE assigned account. Use only the account your trainer gives you.

#	Account (email)	#	Account (email)
1	n8n18oct1@tertiaryinfotech.com	7	n8n18oct7@tertiaryinfotech.com
2	n8n18oct2@tertiaryinfotech.com	8	n8n18oct8@tertiaryinfotech.com
3	n8n18oct3@tertiaryinfotech.com	9	n8n18oct9@tertiaryinfotech.com
4	n8n18oct4@tertiaryinfotech.com	10	n8n18oct10@tertiaryinfotech.com
5	n8n18oct5@tertiaryinfotech.com	11	n8n18oct11@tertiaryinfotech.com
6	n8n18oct6@tertiaryinfotech.com	12	n8n18oct12@tertiaryinfotech.com

Password: provided by your trainer in class · Do not change the password or save real credentials in it.

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TOPIC 1 · SIGN IN

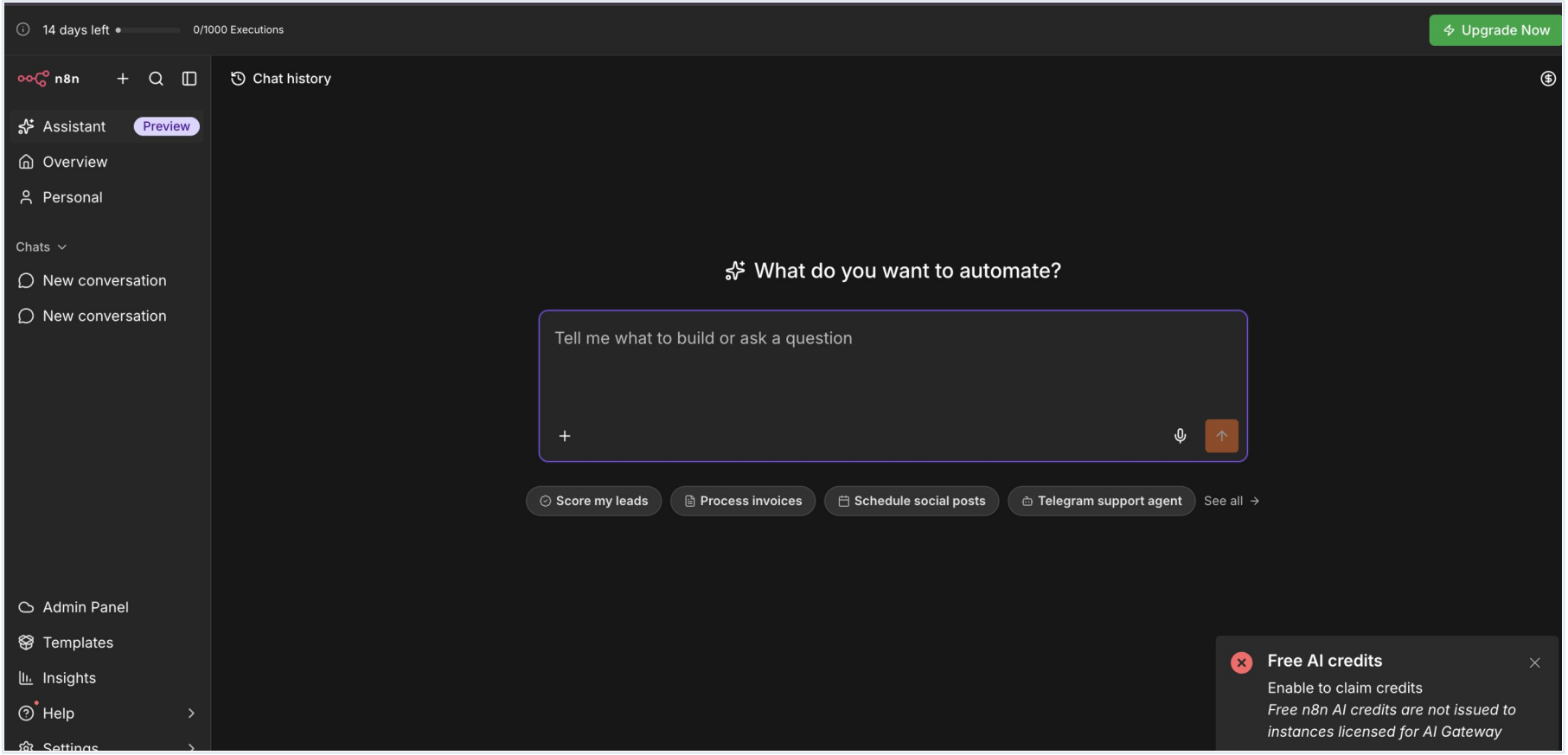
Sign In to n8n

1. Go to app.n8n.cloud/login — or open your own instance URL directly
2. Enter your assigned account email, e.g. n8n18oct5@tertiaryinfotech.com
3. Enter the password your trainer gives you
4. Open your instance from the dashboard — a free 14-day trial, active through the training
5. You land in n8n — open Assistant or Overview to start building

Your account is already set up — the name is your email prefix and onboarding is done. Do not change the password.

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n8n Assistant — Describe It, n8n Builds It



Open Assistant in the left sidebar and answer "What do you want to automate?". It plans, builds and tests a normal n8n workflow you can inspect, edit and publish. Assistant chats use the plan's monthly Assistant credits.

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TOPIC 1 · DEMO

Demo: Build a Social Media Campaign Workflow with n8n Assistant

Give n8n Assistant one clear prompt — it plans and builds the workflow below, with a human approval gate.

Form Trigger
you enter a topic

▶

Research
AI Agent + web search

▶

Draft post
AI writes the copy

▶

Human in the loop
email approve / reject

▶

Schedule post
wait, then publish

THE PROMPT

Create a workflow for a social media marketing campaign:
1. Start from a form where I enter a campaign topic.
2. Research the topic on the web and summarise 3 key points.
3. Draft a LinkedIn post under 150 words from the research.
4. Email me the draft and wait for my approval before posting.
5. If approved, schedule the post for 9am the next weekday;
   if rejected, log my feedback in a Google Sheet.

Use Gateway credits for the AI model. Ask me before publishing.

THEN YOU

- Review the plan before it builds
- Connect Gmail and the social account when it asks
- Test with a real topic and approve the draft
- Publish only when the run looks right

The AI built it; the workflow runs it — and nothing is posted until a person approves the draft.

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TOPIC 1 · n8n FUNDAMENTALS

Trigger Nodes — What Is Allowed to Start a Workflow

The trigger you choose is a statement about who or what may start the process.

1 Manual Trigger

Click Execute workflow. For building and testing, and for one-off jobs such as the Lab 5 ingestion run.

2 Schedule Trigger

A clock starts the workflow — an interval or a cron expression. Nightly reports, reminders and syncs.

3 Form Trigger

n8n hosts a web form for you and each submission starts a run. This is Lab 1's front door.

4 Webhook

An external system calls a URL you expose. Your own website, app or CRM — Labs 2, 3, 4 and 5.

5 Chat Trigger

Each message in n8n's built-in chat window starts a run — the quickest way to talk to an agent while building.

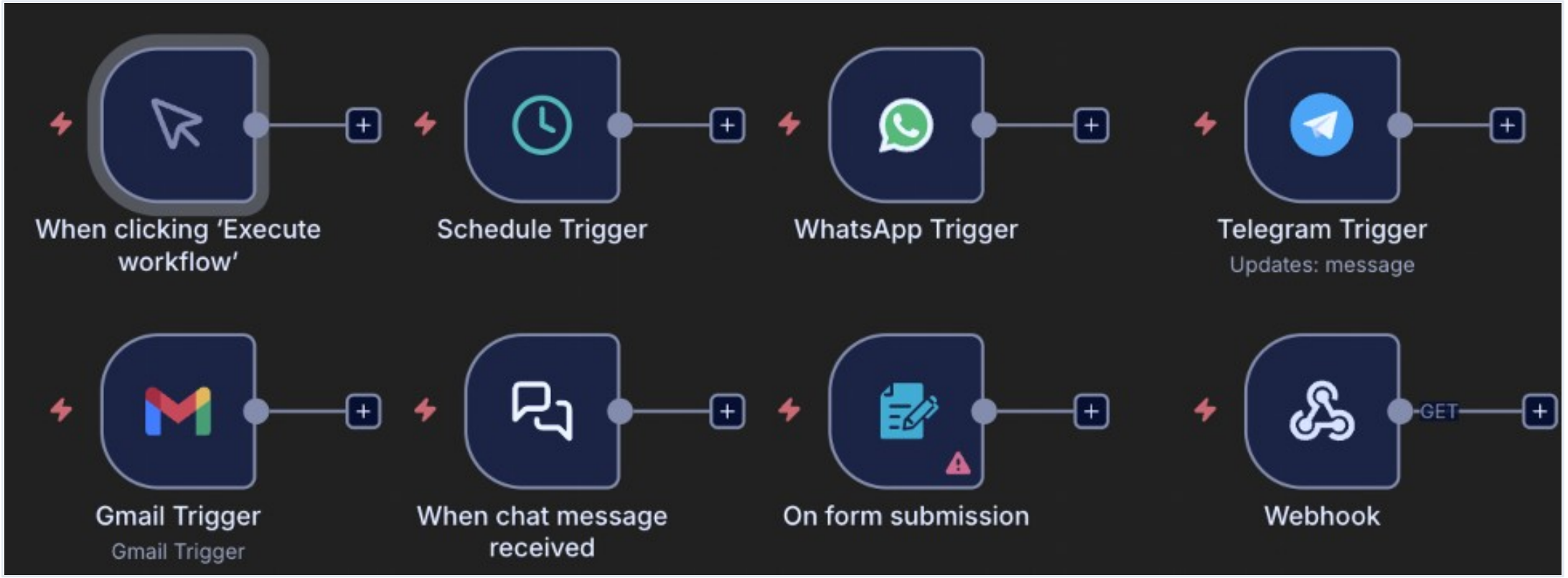
6 App & Email Triggers

On new Gmail, new Sheets row, new Drive file. React to changes in the apps you already use.

Pick the trigger by asking: who or what should be allowed to start this process?

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Trigger Nodes on the n8n Canvas



n8n trigger nodes: Manual · Schedule · WhatsApp · Telegram · Gmail · Chat · Form · Webhook

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Triggers Used in This Course

Every lab's trigger, and why that trigger is the right front door for that business process.

Trigger	Where	Why it fits
Form Trigger	Lab 1	A visitor's event sign-up must start the workflow
Webhook — POST	Labs 2, 3, 4, 5	A website sends JSON and waits for the agent's reply
Manual Trigger	Lab 5 ingestion	A human loads the documents once, by hand
Schedule Trigger	Not used in class	Nightly reports and reminders — same pattern, a clock instead
Chat Trigger	Optional while building	Talk to an agent without building a website first

Swap the front door and keep the workflow — the trigger is the only part that has to change.

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Data-Processing Nodes — The Workhorses

Between the trigger and the output, these nodes shape, route and transform the data.

1 Edit Fields (Set)

Add, rename or reshape fields — the node you reach for first when the data is the wrong shape. Used in Labs 2 and 4.

2 IF / Switch

Branch the workflow. IF gives a true and a false path; Switch gives one path per case. Lab 4 routes Approve vs Decline.

3 Filter

Let only the matching items continue and drop the rest of the list.

4 Merge

Combine two branches — append the rows, match them by key, or choose one side.

5 Split Out / Aggregate

Turn one item holding a list into many items — and back again.

6 Code

JavaScript or Python for the 5% a built-in node cannot do. Lab 3 uses it to normalise the chat message.

7 Loop / Wait

Process items in batches, or pause a workflow for a period of time or an event.

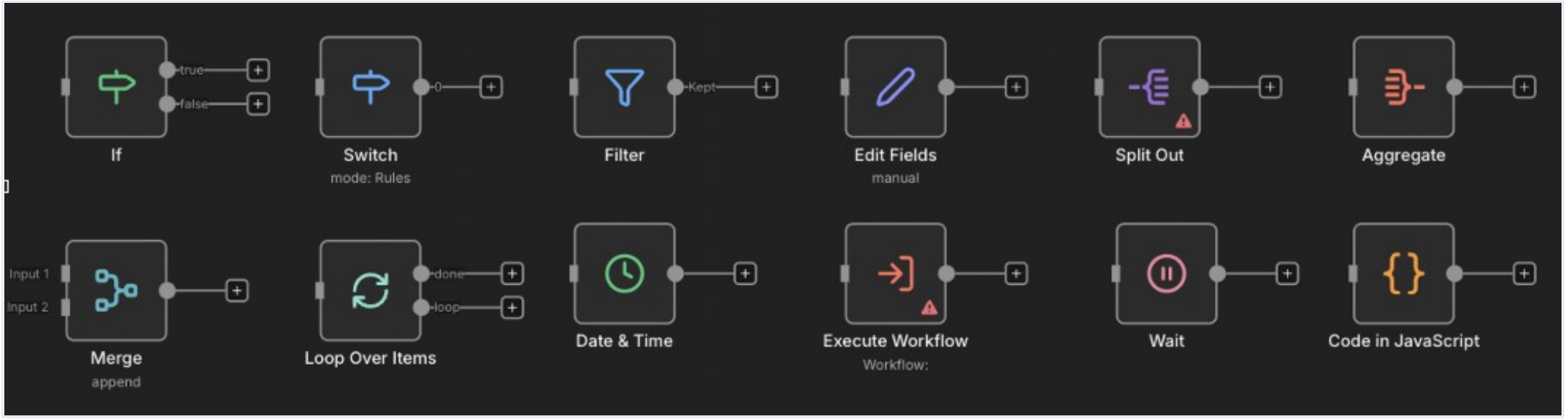
8 HTTP Request

Call any API on the internet — the universal escape hatch when no integration exists.

Most workflows are: trigger → reshape (Set) → decide (IF) → act (app node). Reach for Code last.

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Data-Processing Nodes on the Canvas



If · Switch · Filter · Edit Fields · Split Out · Aggregate · Merge · Loop · Date & Time · Wait · Code

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Where the Data Goes — Storage & App Actions

Action nodes that write the result somewhere useful, or send it to a person.

1 Google Sheets

A real spreadsheet you keep, connected with OAuth2. Append Row is the workhorse — Labs 2 and 4 log every decision here.

2 Gmail / Outlook

Send a message — and in Lab 4, Send and Wait for Response, which turns email into an approval gate.

3 Vector Store

Store embeddings for RAG. In-memory for class, Pinecone / Supabase / Qdrant for production — Lab 5.

4 Respond to Webhook

Return data to whoever called the webhook, so a web page can render the agent's answer.

Anything you must keep goes to external storage — an in-memory store vanishes when n8n restarts.

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JSON and Expressions — How Data Moves

Every node passes a list of items; each item is a JSON object of key/value pairs.

- JSON (JavaScript Object Notation) is a simple text format for data.
- Values can be text, numbers, true/false, lists [ ] or nested objects { }.
- Most APIs send and receive JSON — and so does every n8n node.
- Click any node to inspect the JSON going in and coming out.

ONE FORM SUBMISSION AS JSON

```
{
  "Name": "Alice Tan",
  "Email": "alice@corp.com",
  "Phone": "91234567",
  "Message": "Please reserve 2 seats"
}
```

Read any value later with an expression: {{ $json.Name }} → "Alice Tan"

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Test vs Production Executions

The same workflow runs in two modes — always know which one you are looking at.

TEST — in the editor

- Click Execute workflow to run it once.
- The data is shown live on the canvas, node by node.
- Pin a node's output to freeze it while you build downstream.
- Perfect while building — nothing needs to be published.

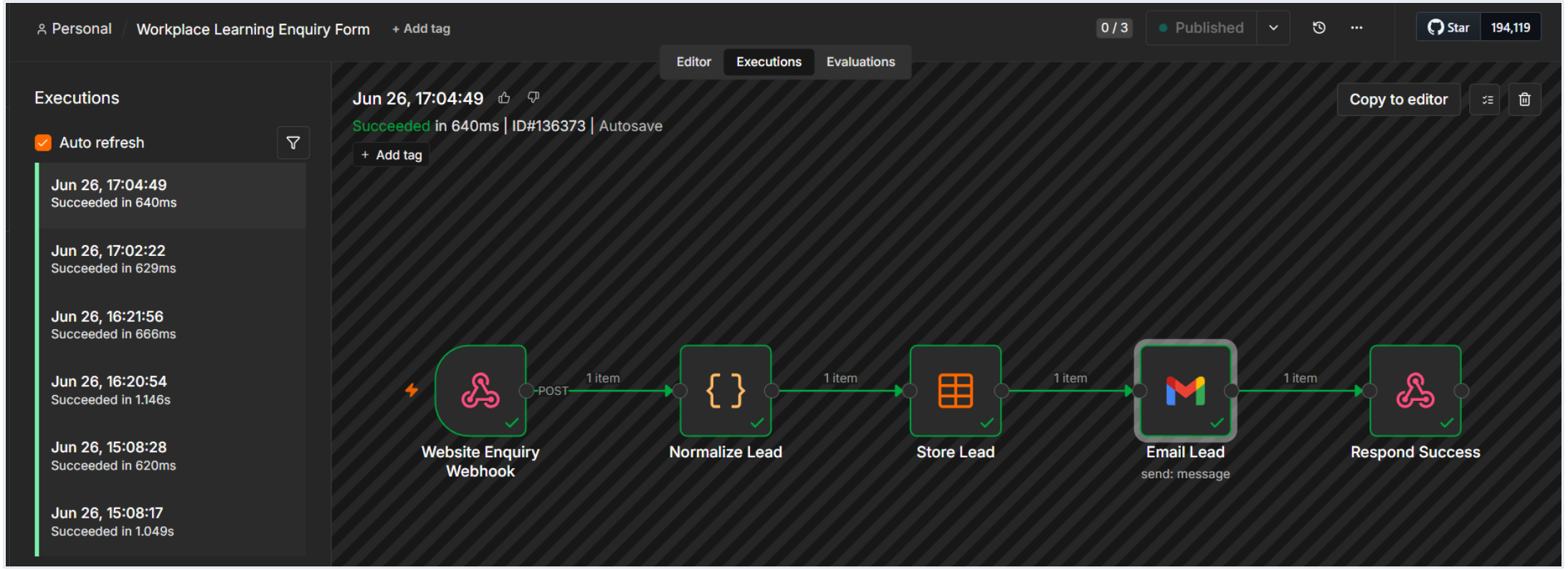
PRODUCTION — after you click Publish

- Runs automatically whenever the trigger fires.
- Nothing appears on the canvas — runs are logged under Executions.
- Every run records status, timing and the full data at each node.
- This is what your users actually experience.

The Executions list is the honest account of what happened — debug from there, not from memory.

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Execution History



The Executions tab — every run shows its status, duration, ID and the full data at each node

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Credentials — Connect Your Accounts Once

Nodes reference credentials by NAME. Set each one up once under Credentials → Add credential.

1 Gmail (OAuth2)

Sign in with Google and grant access. Used from Lab 1 onward for sending mail and for the Lab 4 approval gate.

2 Google Gemini (API key)

The LLM behind every agent in this course, and the embedding model in Lab 5.

3 Google Sheets (OAuth2)

The audit log in Labs 2 and 4 — customer records, decisions, drafts and approvals.

4 Google Drive (OAuth2)

The brochure folder that Lab 5's ingestion workflow reads from.

Imported workflows carry credential NAMES, not secrets — always re-select your own after importing.

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Lab 1 — n8n Trigger and Action (Event Flyer with QR Code)

Your first n8n workflow: a form emails each sign-up to the administrator, then you publish it and put its QR code on an event flyer.

Form Trigger
visitor submits the sign-up

▶

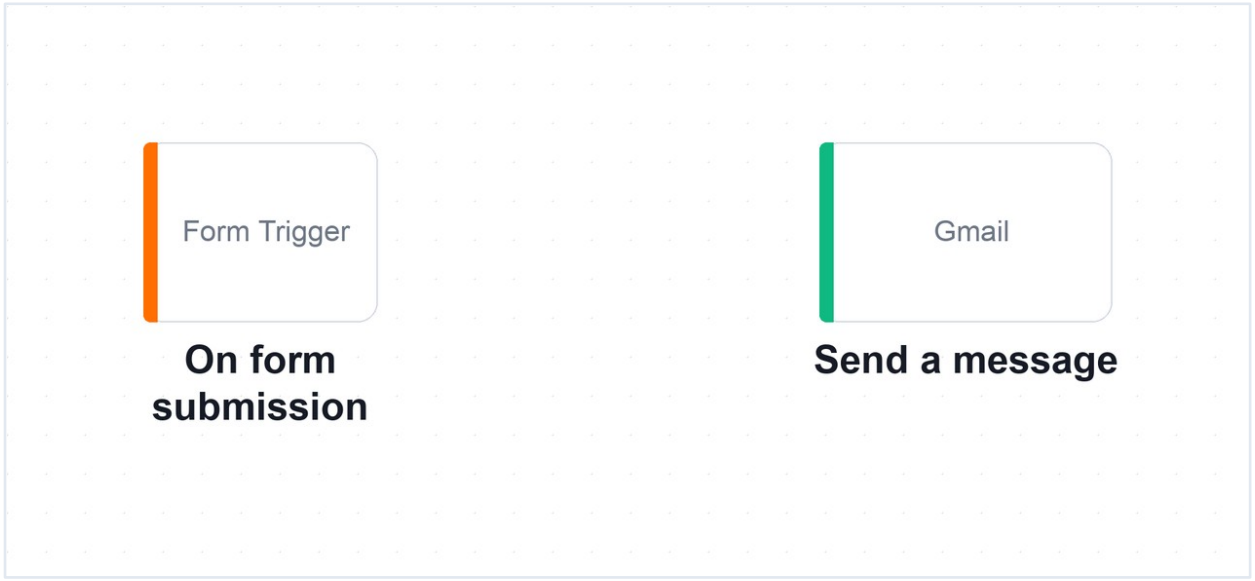
Gmail — Send a message
administrator is emailed

▶

Publish
copy the Production URL

▶

QR code flyer
anyone scans to sign up



The Lab 1 workflow on the n8n canvas. Two nodes: the Form Trigger decides WHEN the workflow runs, and the Gmail node decides WHAT it does.

LAB 1

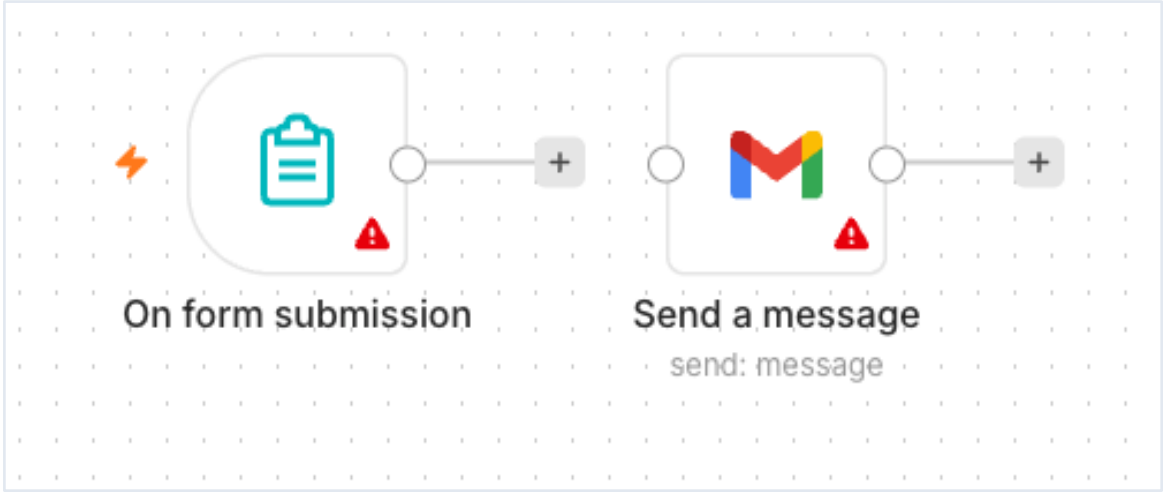
n8n Trigger and Action (Event Flyer with QR Code)

Form Trigger → Gmail (Send a message) → Publish → QR-code flyer · full click-by-click steps in the Learner Guide

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Lab 1 — n8n Trigger and Action — What You'll Build

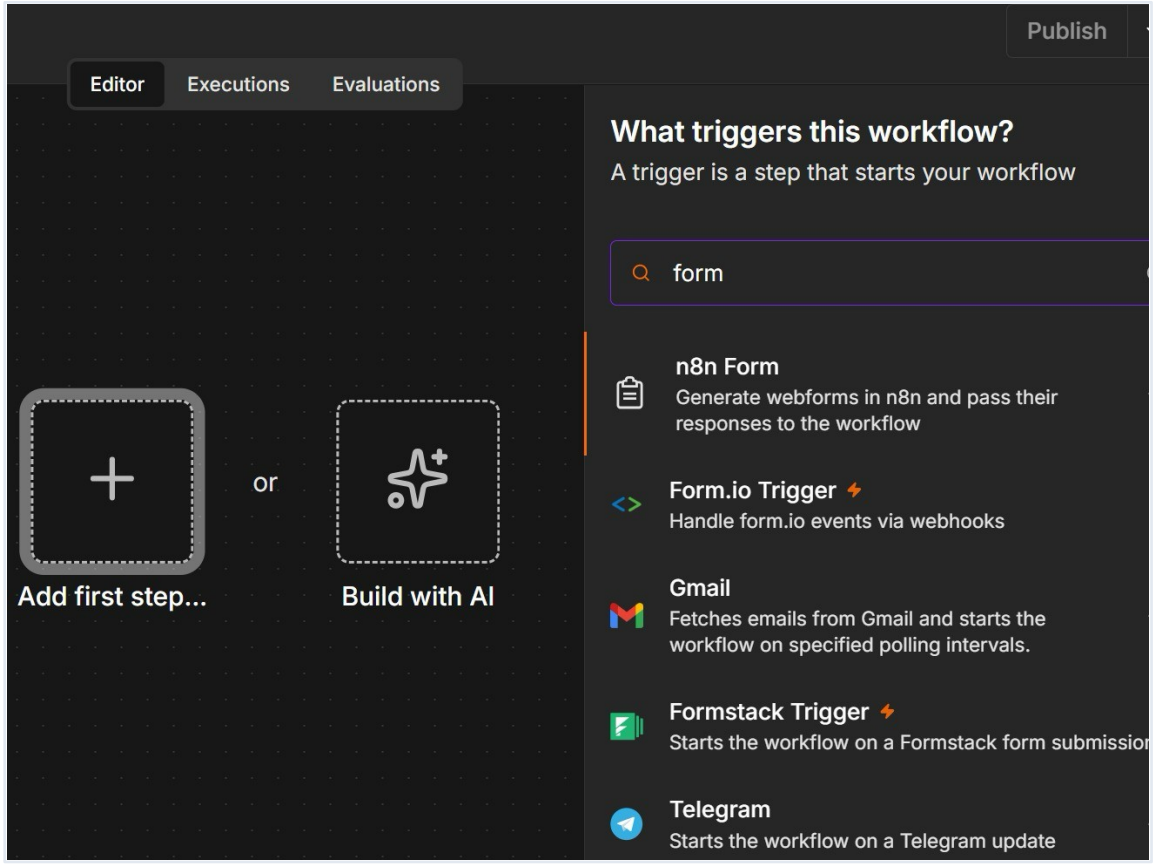
- The finished Lab 1 workflow on the n8n canvas — the Form Trigger fires on each submission and the Gmail node sends the details to the administrator.
- A trigger decides when a workflow runs; an action decides what it does.
- Expressions such as {{ $json.Name }} carry data from one node to the next.

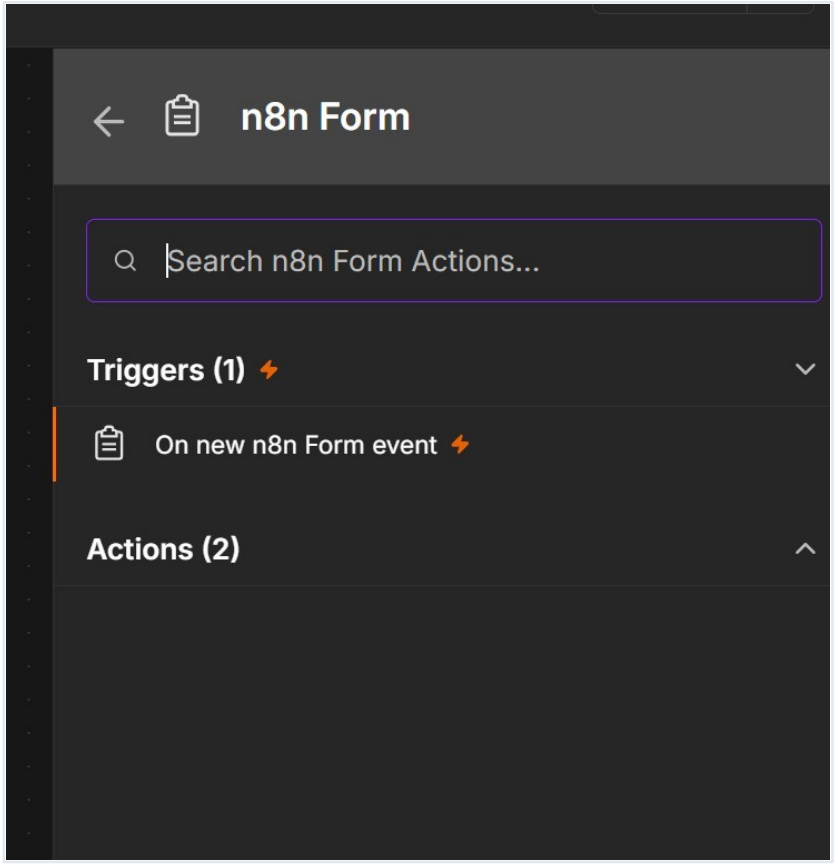


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LAB 1 · STEP BY STEP

Lab 1 — Step 1: Add the n8n Form Trigger



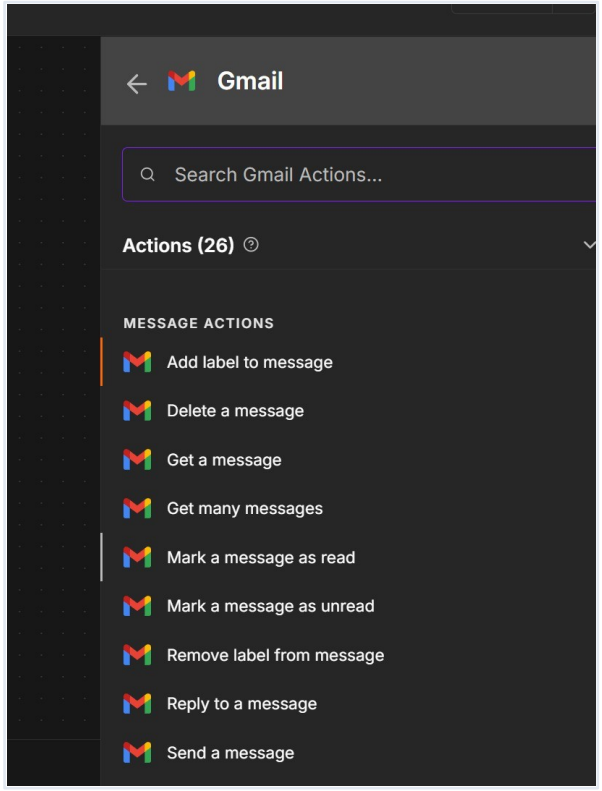


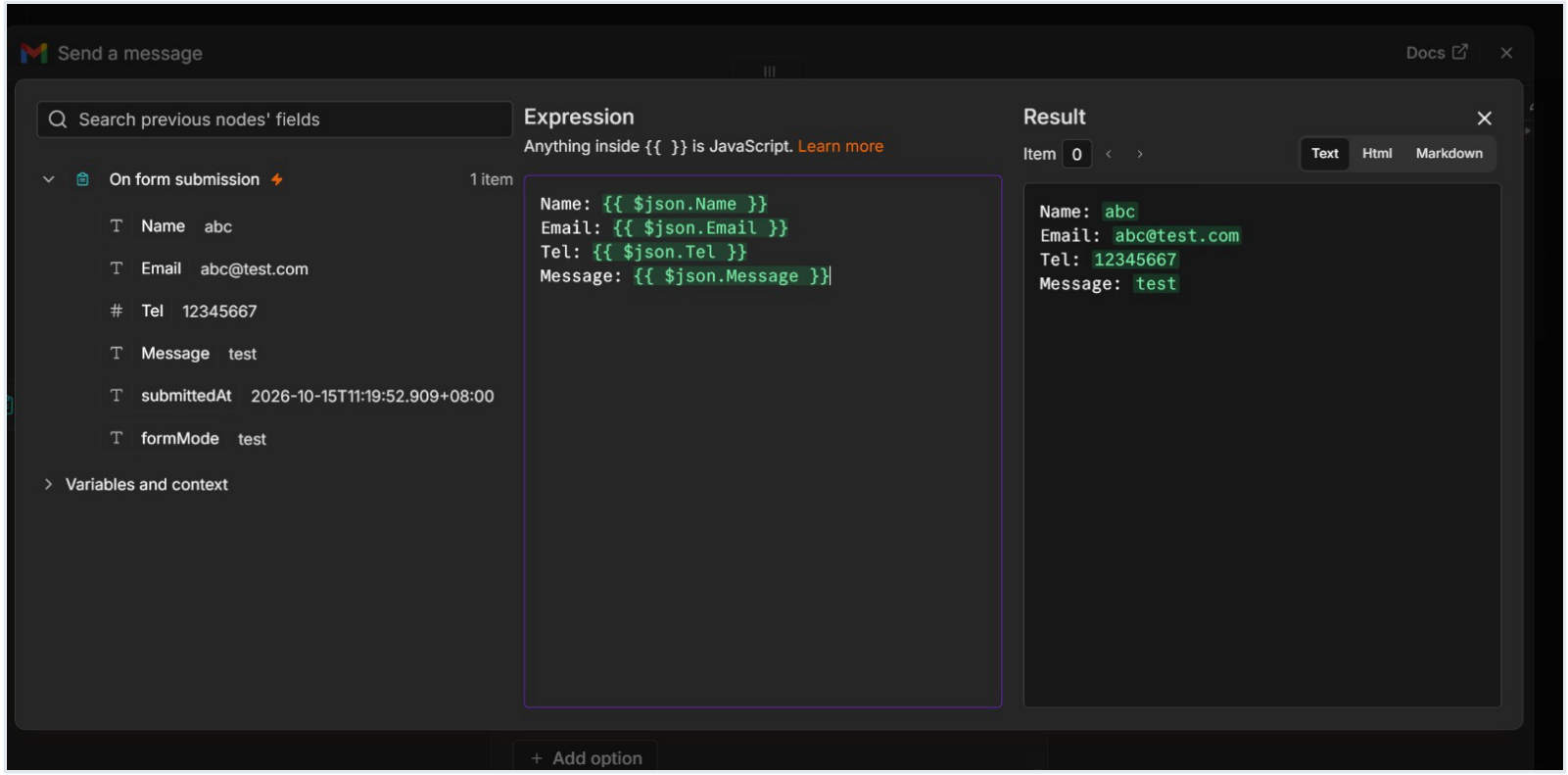
Click Add first step (+), search 'form' and choose n8n Form → On new n8n Form event. Then add the Name, Email, Phone and Message fields.

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LAB 1 · STEP BY STEP

Lab 1 — Step 2: Gmail — Send a Message with Expressions





Add Gmail → Send a message. Switch Message to Expression and drag in the form fields — the Result panel previews the email before you run it.

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LAB 1 · STEP BY STEP

Lab 1 — Step 3: Publish and Copy the Production URL

1. Test first: Execute workflow, submit the test form, check the email arrives
2. Click Publish (top right of the editor) — the workflow is now live
3. Open the Form Trigger node
4. Switch the form URL from Test URL to Production URL
5. Copy the Production URL — this is what the QR code will open

The Test URL answers one submission while you watch; a PUBLISHED workflow's Production URL works for everyone.

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LAB 1 · STEP BY STEP

Lab 1 — Step 4: Create a Flyer with the QR Code

1. Paste the Production URL into alfredang.github.io/qrcodegenerator
2. Download the QR code as a PNG
3. Design the flyer — Canva, PowerPoint or an AI image tool
4. Add the QR code with a clear 'Scan to register' call-to-action
5. Present it — classmates scan, register, and the admin is emailed

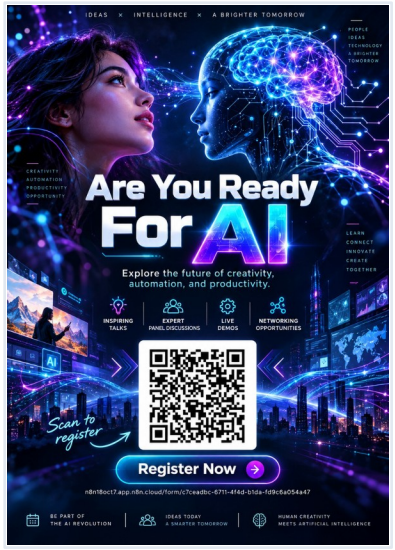
A two-node workflow is now a real sign-up channel: anyone with a phone can register by scanning your flyer.

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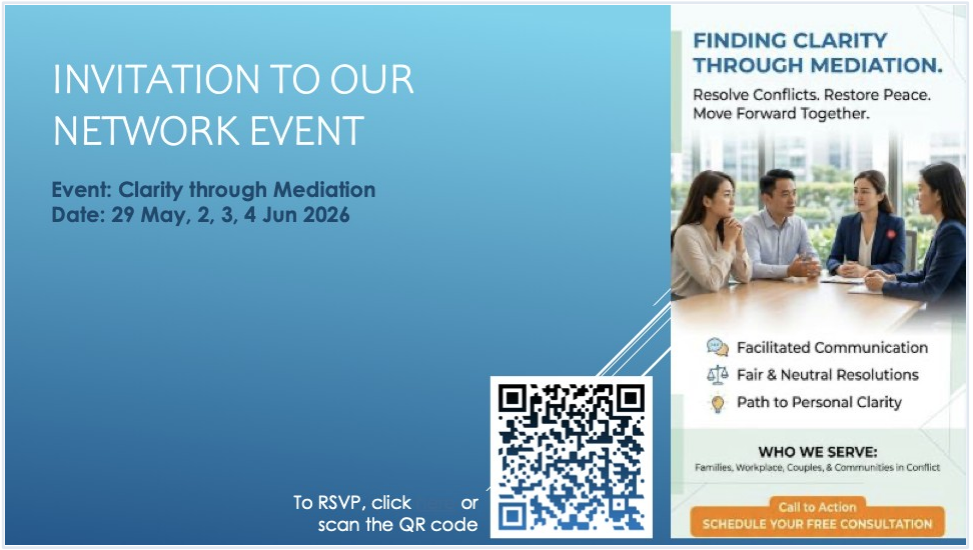
LAB 1 · INSPIRATION

Lab 1 — Flyers from Previous Learners







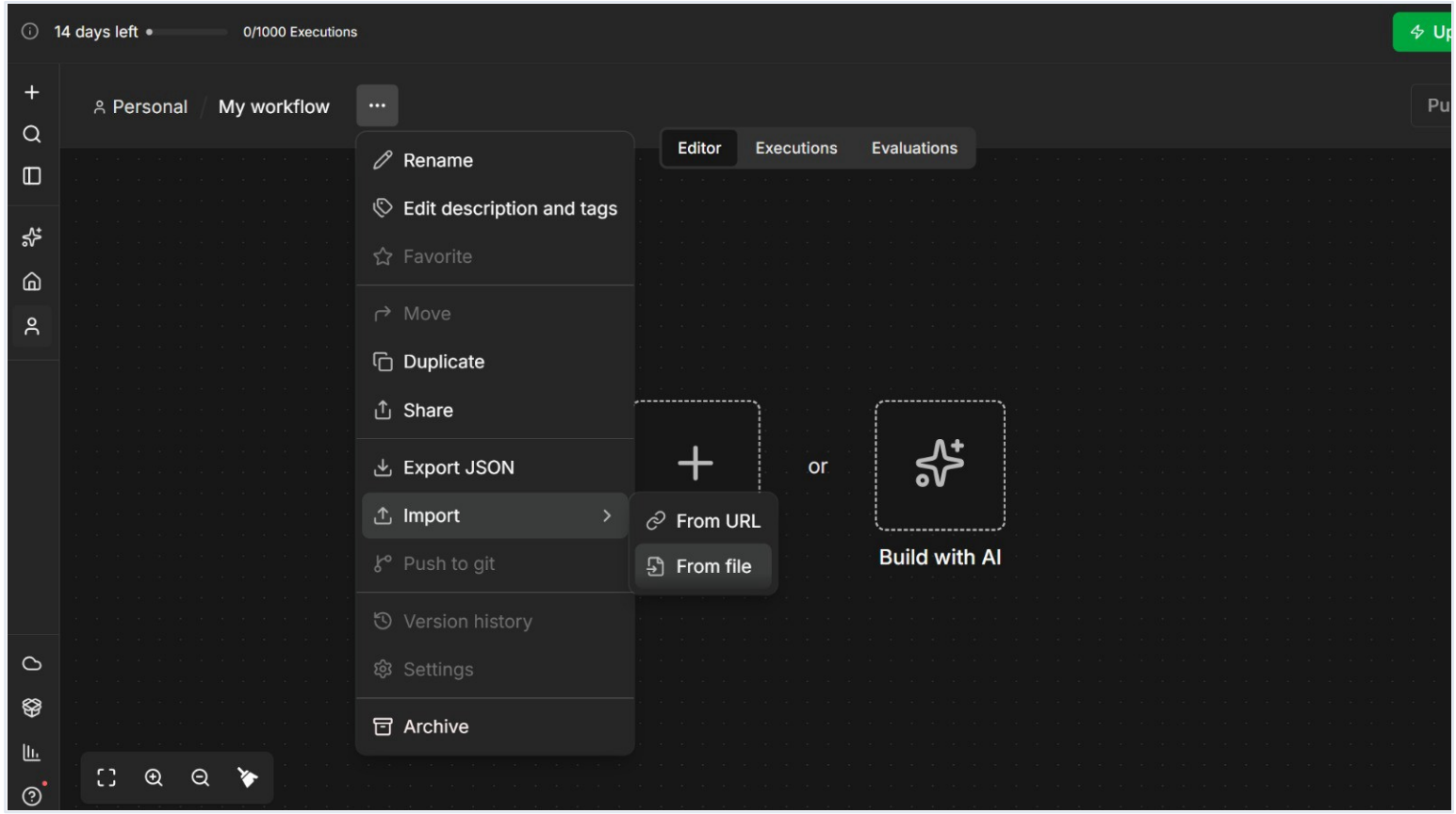


Each QR code encodes the form's Production URL — scan it, register, and the event administrator receives the details by email.

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LAB 1 · STEP BY STEP

Lab 1 — Compare with the Reference Solution



Create a NEW workflow, open the '...' menu → Import → From File and pick Lab 1 - Event Flyer with QR Code.json, so your own build is not overwritten.

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LAB 1 · VERIFY AND DEBRIEF

Lab 1 — n8n Trigger and Action

Scan the QR code on your flyer with your phone, submit the form, and confirm the administrator's inbox receives the email with the details you typed. A reference solution is provided as labs/lab1-trigger-actions/Lab 1 - Event Flyer with QR Code.json — build yours by hand first, then import it to compare.

1 Triggers and actions

A trigger decides when a workflow runs; an action decides what it does.

2 Expressions

Expressions such as {{ $json.Name }} carry data from one node to the next.

3 Test vs Production URL

The Test URL works only while you are listening; the Production URL works once the workflow is published.

Lab 1 — you can now build an automated workflow in n8n using a trigger and an action.

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Lunch Break

1 hour

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TOPIC 1 · AI AGENT BUILDING BLOCKS

Before Lab 2 — The Building Blocks of an AI Agent

Lab 1 was a trigger and an action. Lab 2 adds an AI Agent that decides — these six blocks are what it is made of.

AI Agent
the hub that decides

▶

Model
the LLM brain

▶

Tools & MCP
what it may do

▶

Instructions
the rules it follows

▶

Memory
what it remembers

▶

Webhook
how the world calls it

1 What Lab 1 had

A Form Trigger and a Gmail action. Every submission follows the same fixed path — no judgement anywhere.

2 What Lab 2 adds

An AI Agent node with a model, three tools and the bank's rules, deciding APPROVED, REJECTED, REVIEW or DUPLICATE.

3 What changes at the front door

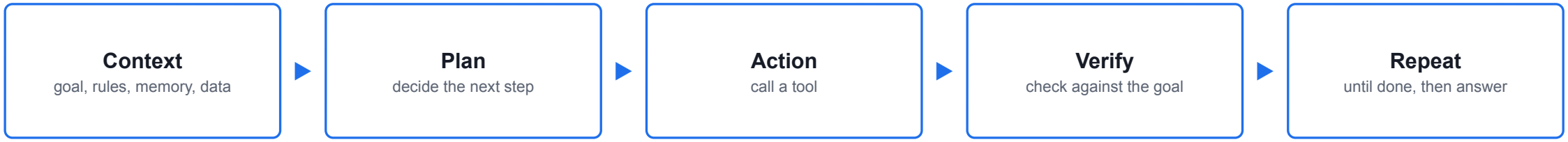
The n8n form is replaced by a real bank website that POSTs to a Webhook and reads the decision back.

Learn the six blocks one at a time, then watch them work together in Lab 2.

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What Is an AI Agent? — The Agent Loop

Given a goal, an agent does not answer once. It works in a loop until the goal is met.



1 · Context

Everything the agent can see: the goal, its instructions, memory of the conversation, and what earlier tool calls returned.

2 · Plan

The LLM reasons over the context and chooses the next step — which tool to call, with what input, or whether it is finished.

3 · Action

The agent calls a tool: look up a record, write a row, send an email. This is where it acts on the world.

4 · Verify and repeat

The tool's result goes back into the context and is checked against the goal. Not done? Plan again. Done? Reply.

Every turn of the loop is an LLM call — so cap it (Max Iterations on the AI Agent node) and log every step.

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Components of an AI Agent

Four parts around one hub: the AI Agent node wires them together, and every agent in this course is exactly these four.

MODEL

The LLM 'brain' — interprets language, reasons and chooses the next step. Google Gemini in this course.

MEMORY

Keeps the conversation so follow-up questions still make sense. Keyed on a session ID, per visitor.

THE AI AGENT

plans · decides · calls tools

TOOLS

The abilities it may use: look up a Sheets row, search a vector store, send an email, call an API.

INSTRUCTIONS

The system message — identity, scope, rules and refusals. Always in force, on every single turn.

Model + Memory + Tools + Instructions. Change any one of the four and you have a different agent.

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The Model — Popular LLM Providers

Any chat model plugs into the same AI Agent node. n8n lists each provider's live models — swap the brain, keep the agent.

Provider	Models	Notes
Google	Gemini 4 Argon · Gemini 3.8 Flash · Gemini 3.1 Pro	Course default (Gemini Flash) — also supplies the embeddings in Lab 5
OpenAI	GPT-6 Astra · GPT-6.1 Sol · GPT-6 Luna	Newest family; GPT-5.x models remain in the API
Anthropic	Claude Opus 5.5 · Sonnet 5.5 · Haiku 5.5	Strong reasoning and long context; Haiku is fast and cheap
Meta	Llama 4 (Scout, Maverick)	Open weights — can be self-hosted, e.g. with Ollama
Mistral	Mistral Large 4 · Mistral Medium 3.5	A European option, with open-weight models too

Pick by task: small fast models for classification and guardrails, larger models for reasoning.

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Model Parameters — Temperature and Friends

Settings on the chat-model node shape HOW the model answers. The most important by far is temperature.

1 Temperature 0 – 0.3

Deterministic and repeatable. Use for classification, compliance decisions and data extraction — automation that must give the same answer twice.

2 Temperature ≈ 0.7

Balanced. Natural-sounding conversation with some variety — the usual default for a customer-facing assistant.

3 Temperature 1.0 +

Creative and surprising. Marketing copy and brainstorming — never for a decision that must be defended.

1 Top-p

An alternative sampling control that limits the model to the most probable tokens. Tune temperature OR top-p, never both.

2 Max tokens

Caps the length of the reply. Protects your cost and keeps chat answers short enough to read.

3 Frequency / presence penalty

Discourage repetition and nudge the model onto new topics — useful for long generations, rarely for automation.

For business automation keep the temperature LOW — Lab 3 uses 0.2 so the same question is refused the same way every time.

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Claim Your Free AI Credits

1. Sign in to n8n Cloud with your training account — the free trial includes Gateway credits
2. Open a workflow and add an AI node, e.g. AI Agent → Gemini or OpenAI Chat Model
3. In the node's Credential field, choose Use Gateway credits — your balance shows beside it
4. Pick a supported model and run the node — no provider account or API key needed
5. Track the balance on the node, or in Admin Panel → Gateway credits

A red "Free AI credits — Enable to claim credits" toast is the legacy OpenAI offer: Gateway instances don't get it. Use Gateway credits.

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Tools — What the Agent May Do

Attach tools to the agent's Tool input. The model decides WHEN to call one, based on the description you write.

1 Google Sheets tool

Look up or write structured records. Lab 2 attaches three: check for a duplicate, create the customer, send the confirmation.

2 Vector Store tool

Retrieve document chunks for a grounded answer — the Query Data Tool in Lab 5.

3 HTTP Request tool

Call any external API in the middle of a conversation — prices, stock levels, a CRM record.

4 Sub-workflow tool

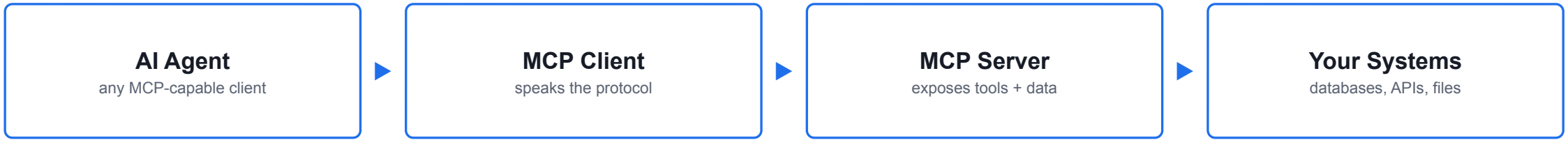
Wrap an entire n8n workflow as one callable tool, so your own logic becomes something the agent can invoke.

The tool DESCRIPTION is how the agent knows when to call it — write it for the model, not for a developer.

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MCP — Model Context Protocol

An open standard that gives agents a common plug for tools and data, instead of one custom integration per pair.



Without MCP

Every agent × every tool needs its own custom integration. N agents and M tools means N × M connectors to build and to maintain.

With MCP

Build ONE MCP server per system and any MCP-capable agent can use it. N + M pieces, and the tools become reusable across agents.

MCP is like USB-C for AI tools — one connector, many devices.

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MCP in n8n

n8n sits on both sides of the protocol — it can serve tools to other agents, and borrow tools for its own.

MCP SERVER TRIGGER — n8n serves tools

- Expose your own n8n workflows AS tools over MCP.
- Claude, an IDE or another agent can then call your workflows directly.
- n8n becomes the tool layer for any MCP client.
- Secured with a bearer token on the server URL.

MCP CLIENT TOOL — n8n consumes tools

- Attach an external MCP server to your n8n AI Agent.
- The agent gains all of that server's tools with no custom node.
- Mix MCP tools and native n8n tools freely on the same agent.
- Same agent loop — simply more reach.

Two directions, one protocol. Build the tool once and every agent you own can use it.

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Instructions — The Four Parts That Constrain

The system message is always in force. Every good agent instruction in this course states four things.

1 Identity

Who the agent is, and for whom. 'You are the onboarding assistant for Marina Trust Bank.' Narrow roles refuse far more reliably than general assistants.

2 Source rules

Where answers may come from. 'Answer ONLY from the brochures returned by the tool — never from general knowledge.'

3 Refusals

What is out of bounds, stated explicitly. 'Never recommend a product, predict a return, or guarantee an outcome.'

4 Escalation

What to do at the boundary. 'If you are not sure, say so and offer a consultation with a licensed adviser.'

Vague instructions produce confident wrong answers — every rule you skip, the model improvises.

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Examples of Good and Bad Instructions

The same chatbot, two system messages — one is testable, the other is a hope.

✗ TOO VAGUE

"You are a helpful assistant. Answer the user's questions politely."

No identity · no source rules · no refusals · no escalation.

It will happily invent a course fee and guarantee an investment return — politely.

✓ CONSTRAINED AND TESTABLE

"You are the enquiry assistant for Meridian Asset Management.

Answer only questions about our services, fees and appointment booking.

Use the brochure search tool for any factual question — never answer from general knowledge.

Never recommend a product, predict a return, or guarantee an outcome.

If you are not sure, say so and offer a call with a licensed adviser."

Identity · source rules · refusals · escalation — you can test each line by trying to break it.

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Memory — Why the Agent Remembers

Without memory every message is from a stranger; with it, 'what about next month?' still makes sense.

Msg 1: 'What courses do you run?'
agent answers from its tools

▶

Stored in memory
keyed by session ID

▶

Msg 2: 'How much is that one?'
agent knows which course

▶

Window slides
only the last N turns are kept

1 Window buffer memory

Keeps the last N exchanges of THIS conversation. It is n8n's default and it is what Lab 3's chatbot uses.

2 Session key

Memory is per session ID, so two visitors on the same website never see each other's conversation.

3 Sometimes you want NO memory

Lab 2's onboarding agent deliberately has none — one applicant's details must never influence the next decision.

Memory is conversation context, NOT long-term knowledge. Facts belong in tools and documents (RAG).

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Structured Output — Making an Agent's Answer Usable

By default an agent returns a paragraph of prose. Prose cannot be branched on, sorted, counted or coloured.

1 Attach an Output Parser

A Structured Output Parser forces named fields instead of free text — Labs 2 and 4 both use one.

2 Name the fields you will act on

Lab 2 returns decision, reason and riskFlags. The website colours its result card from decision alone.

3 Enumerate the allowed values

APPROVED / REJECTED / REVIEW / DUPLICATE. A closed list is what makes an IF node reliable.

4 Everything downstream depends on it

An IF node routes on it, a Sheets node logs it, and a queue is sorted by it. Prose supports none of that.

If something downstream must act on the answer, make the answer a field — not a sentence.

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TOPIC 1 · WEBHOOKS

What Is a Webhook?

A URL your workflow listens on. When the outside world calls it, your workflow runs — the reverse of you calling an API.

Event happens
a visitor submits the form

▶

HTTP request to your URL
the website POSTs JSON

▶

Workflow runs
n8n receives the data as its input

▶

Respond to Webhook
your reply returns to the caller

1 Push, not pull

No polling and no schedule — the caller notifies you the moment something happens.

2 Any caller

Websites, payment providers, forms, CRMs, mobile apps — anything that can send an HTTP request.

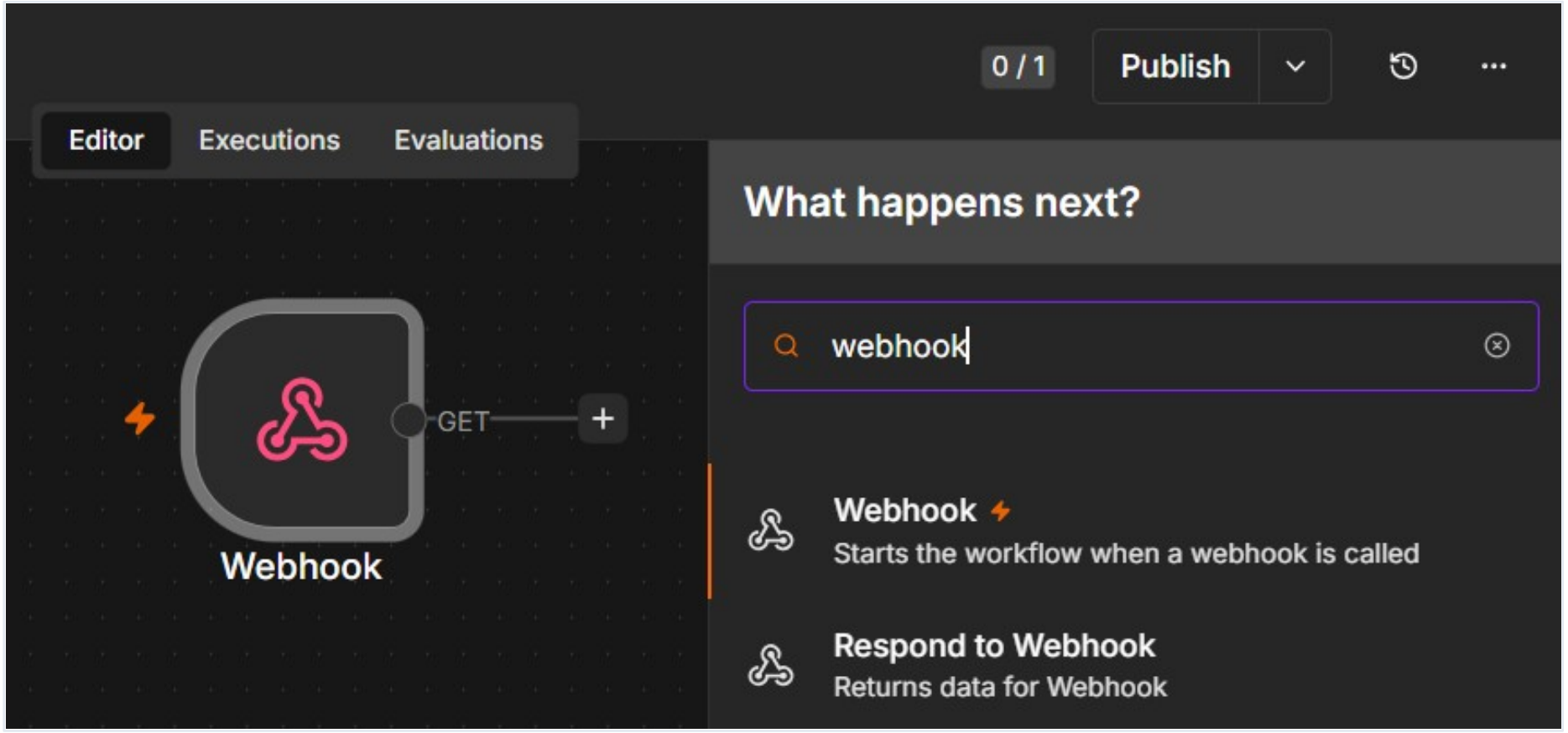
3 A paired node

The Webhook node starts the workflow; Respond to Webhook sends the answer back on the same connection.

A webhook is the reverse of an API call — the outside world calls YOU when an event happens.

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Webhook Nodes in n8n



The Webhook node starts the workflow when it is called; Respond to Webhook returns data to the caller

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TOPIC 1 · WEBHOOKS

Webhook vs HTTP Request — Which Way the Call Goes

Both are plain HTTP. The only difference is who dials.

WEBHOOK — inbound

- An external service calls YOUR workflow.
- You react the moment the event happens.
- It is the TRIGGER — always the first node.
- Labs 2–5: the lab website calls your agent.

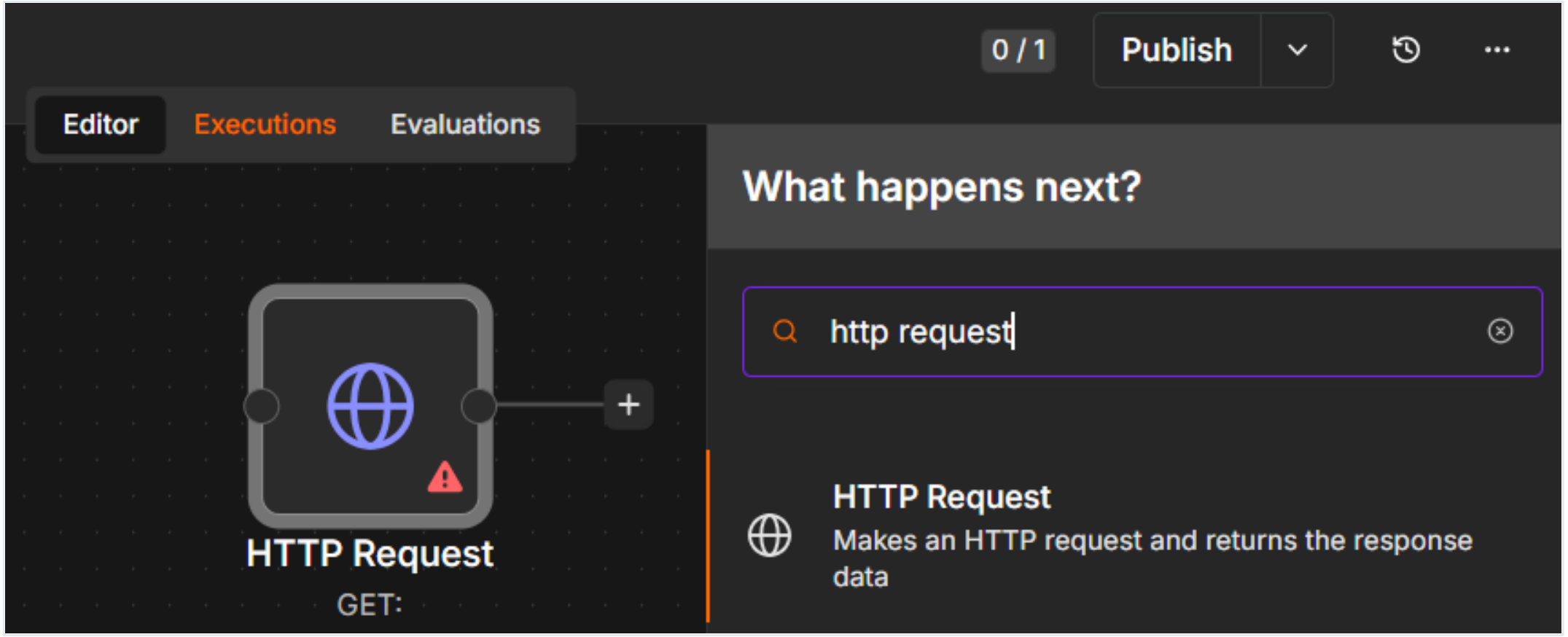
HTTP REQUEST — outbound

- YOUR workflow calls an external service.
- You pull the data at a moment you choose.
- It is an ACTION inside the workflow.
- Use it to reach any API with no dedicated n8n node.

Same HTTP underneath — a webhook receives a call, an HTTP Request makes one.

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The HTTP Request Node



The HTTP Request node calls out to any API — the outbound counterpart to an inbound webhook

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HTTP Methods on a Webhook — GET vs POST

The method is part of the contract with the caller: where the data rides, and what the call means.

GET — read

- The data rides in the URL query string:
  - ?email=alice@corp.com
- A browser can call it just by visiting the URL.
- Read it in n8n as {{ $json.query.email }}.
- Right for: a dashboard pulling data to display.

POST — send

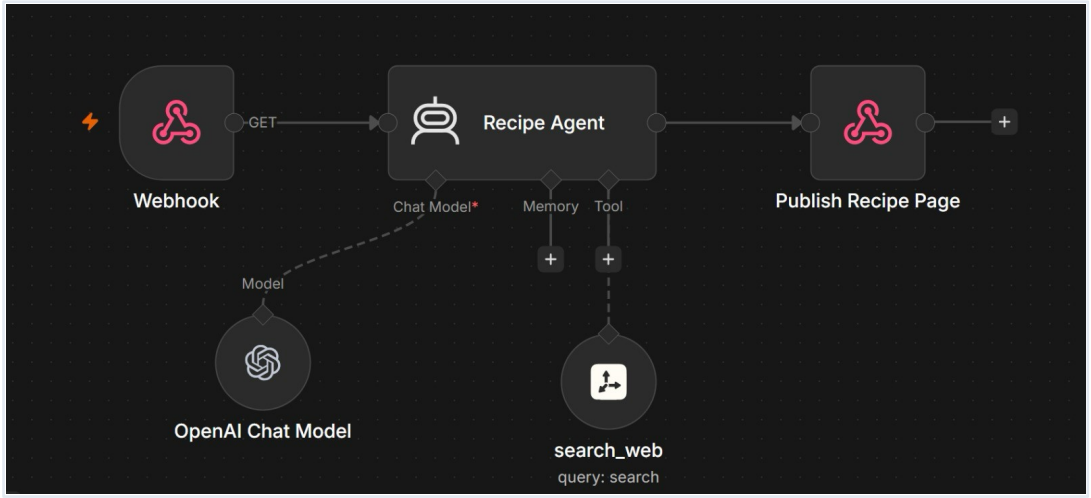
- The data rides in the request body as JSON:
  - { "message": "How much is the course?" }
- This is what a web form or a chat widget sends.
- Read it in n8n as {{ $json.body.message }}.
- Right for: Labs 2–5, where the site submits data.

GET reads, POST sends. Set the method on the Webhook node — and the caller must match it exactly.

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Example: A GET Webhook That Publishes a Recipe Page

One URL, one agent, one page back. A webhook does not have to sit behind a form — opening a link can run an agent.



1 Webhook — GET

Opening the URL in a browser is the request. Anything the agent needs can ride in the query string.

2 Recipe Agent

An OpenAI chat model writes the recipe. There is no memory: each request stands alone.

3 search_web tool

The agent calls search_web, passing a search query, when it needs facts from the web.

4 Page back

Publish Recipe Page (a Respond to Webhook node) returns the page, so the browser shows it at once.

GET suits a link anyone can click. Lab 2's bank form uses POST, because it sends a body of applicant data.

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Webhook Test URL vs Production URL

Every Webhook node gives you TWO URLs. They behave differently, and confusing them is the single most common webhook bug.

TEST URL — /webhook-test/…

- Works only while you have clicked 'Listen for test event'.
- Handles exactly ONE call, then stops listening.
- The run appears live on the canvas — ideal for debugging.
- Dies the moment you leave the editor.

PRODUCTION URL — /webhook/…

- Requires the workflow to be published (Publish, top right).
- Always on — every call starts a run.
- Runs are logged under Executions, not drawn on the canvas.
- This is the URL you give to a website, an app or a QR code.

Your website must call the PRODUCTION URL of a PUBLISHED workflow — the Test URL is spent after one request.

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TOPIC 1 · WEBHOOKS

Configuring the Webhook Node

Four settings decide whether the browser's call actually succeeds.

1 Method + Path

GET or POST, and the path that ends your URL — for example POST /webhook/marina-trust-onboarding.

2 Allowed Origins (CORS)

Browsers block cross-site calls unless you allow them. Set * in class so a local HTML page can call n8n; never leave * in production.

3 Response mode

Immediately (a bare acknowledgement), when the workflow finishes, or via a Respond to Webhook node — every lab here uses the last one.

4 Authentication

None for a classroom demo; Basic, Header or JWT for anything real. A public URL is a public door, and every call costs you model tokens.

'Failed to fetch' in the browser is almost always CORS or a dead Test URL — check those two first.

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LAB 2 · PUTTING IT TOGETHER

Lab 2 — Webhook and AI Agent — How the Building Blocks Fit

Website → Webhook → AI Agent → Respond. Every block you have just learned has one job in this lab.

Building block	In this lab	Why it matters
Webhook	Onboarding Webhook — POST /marina-trust-onboarding	The bank website's front door; Respond to Website returns the decision
AI Agent	Onboarding AI Agent	Applies the bank's onboarding rules and decides
Model	Google Gemini Chat Model (Gateway credits)	The LLM brain — keep the temperature low for repeatable decisions
Tools & MCP	check_duplicate_customer · create_customer_record · send_confirmation_email	The agent chooses when to call each one
Instructions	System message: eligibility rules, age check, four decisions	Where the bank's policy actually lives
Memory	None — deliberately	One applicant must never influence the next decision
Structured output	Decision Parser — decision, reason, riskFlags	Lets the page colour the result card

Six building blocks, one workflow: the webhook brings the applicant in, the agent decides, the page shows the result.

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Lab 2 — Webhook and AI Agent (Marina Trust Bank Onboarding)

Replace the n8n form with a real bank website that POSTs to a webhook, where an AI Agent applies the bank's onboarding rules using three tools.

Webhook — POST
the bank site submits

▶

Normalise
Set node tidies the fields

▶

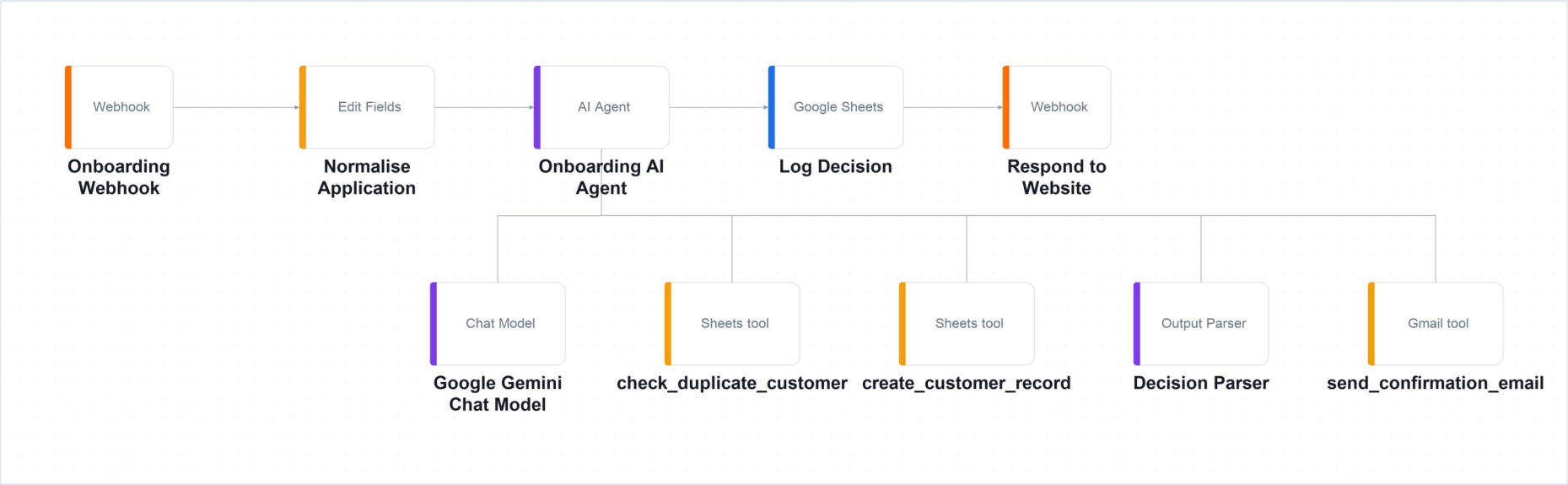
AI Agent
+ 3 tools + parser

▶

Log Decision
Google Sheets audit row

▶

Respond
decision returns to the page



The Lab 2 workflow on the n8n canvas. The main chain runs left to right; the chat model, output parser and three tools hang below the agent on its Tool input.

LAB 2

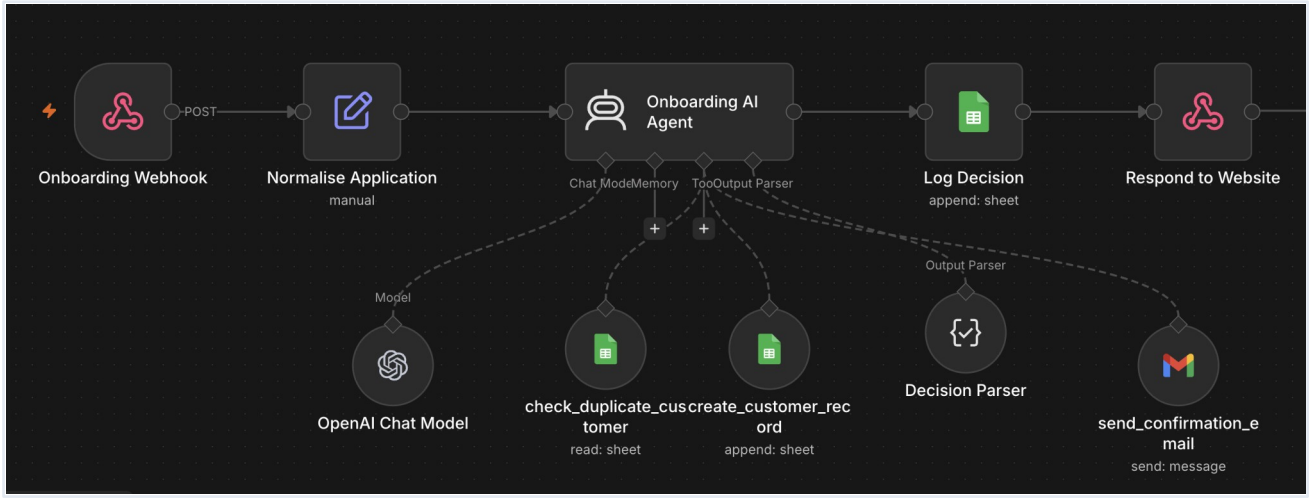
Webhook and AI Agent (Marina Trust Bank Onboarding)

Webhook → Normalise → AI Agent (+3 tools) → Log Decision → Respond to Website · full click-by-click steps in the Learner Guide

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Lab 2 — The Onboarding Workflow in n8n

The imported workflow on the n8n canvas. The main chain runs left to right; the agent's model, tools and parser hang beneath it.



1 Front door

The bank site POSTs to Onboarding Webhook; Normalise Application tidies the fields the agent reads.

2 The agent

Onboarding AI Agent applies the bank's rules with its chat model; Decision Parser returns decision, reason and riskFlags.

3 Three tools

check_duplicate_customer, create_customer_record and send_confirmation_email — the agent chooses when to call each.

4 Log and respond

Log Decision appends an audit row to Google Sheets; Respond to Website sends the decision back to the page.

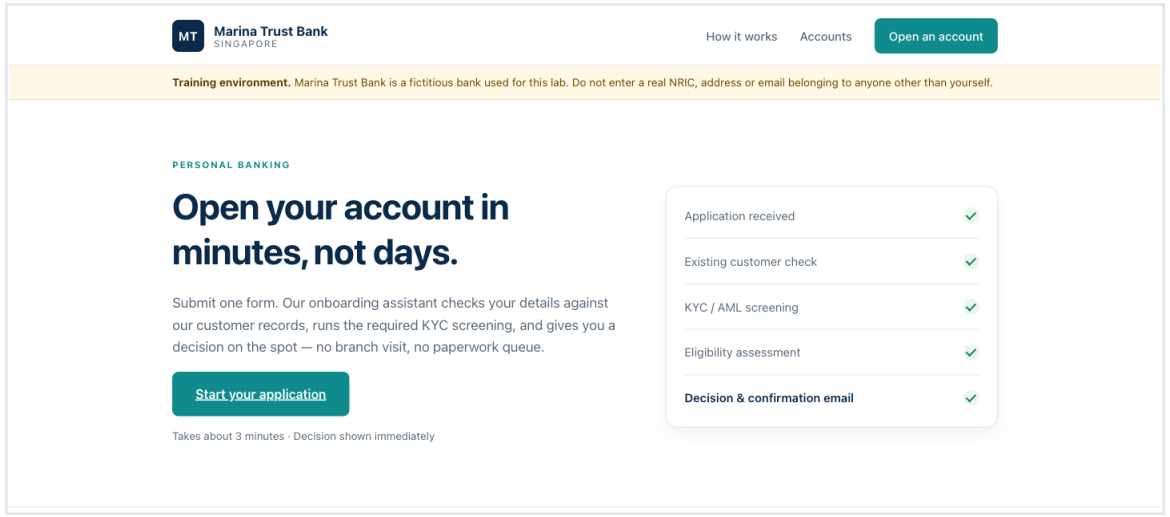
The dashed lines are the agent's sub-nodes: it calls them when it decides to — the solid chain always runs in order.

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LAB 2 · THE APPLICATION

Lab 2 — Webhook and AI Agent — What You'll Build

- The Marina Trust Bank site the webhook sits behind. The applicant submits here, the AI Agent decides, and the page renders the decision as a coloured card.
- A webhook lets any external system — your website, an app, a CRM — start an agent workflow.
- Tools are what turn an LLM into an agent: it decides when to look up a customer and when to create one.



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LAB 2 · VERIFY AND DEBRIEF

Lab 2 — Webhook and AI Agent

Run TC1 (a new applicant) and confirm a green APPROVED card, a new row in the Customers sheet and a confirmation email. Then run TC2 (an already-seeded NRIC) and confirm an amber DUPLICATE card with no new row.

1 Webhooks

A webhook lets any external system — your website, an app, a CRM — start an agent workflow.

2 Tools make an agent

Tools are what turn an LLM into an agent: it decides when to look up a customer and when to create one.

3 Structured output

Structured output (decision, reason, riskFlags) is what lets downstream nodes branch, log and colour a result.

4 The front door is separate

The agent is unchanged from a form-triggered version — only the front door differs.

Lab 2 — you can now deploy an AI agent that is triggered by a webhook from an external website.

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02

TOPIC 02

Agentic Chatbot and Human in the Loop

Agentic workflows · Chatbots on webhooks · Approval gates · Accountability

Applied in Lab 3, Lab 4

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Rule-Based vs Agentic Workflows

In an agentic workflow the LLM makes the routing decisions. You still design the process, the guardrails and the tools.

RULE-BASED (RPA) — every branch wired by hand

- Predictable: the same input always takes the same path.
- Easy to audit — every decision is a rule you can read.
- Breaks on anything you did not foresee.
- Best for fixed, high-volume rules.

AGENTIC — the LLM chooses the path

- Understands free text and intent.
- Handles cases you did not foresee.
- Needs guardrails and logging to stay safe.
- Best for varied, language-driven work — like a chatbot.

What you hand over is the judgement about which path a case takes — never the design of the process.

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TOPIC 2 · AGENTIC CHATBOT

Designing a Safe Agentic Chatbot

Six design rules that turn a helpful chatbot into one a regulated firm can put on its website. This is Lab 3.

1 Define the role narrowly

A chatbot built for one job refuses far more reliably than a general assistant.

2 Write the prohibitions explicitly

Never recommend a product, predict a return or guarantee an outcome — spelled out, not implied.

3 Add a fallback sentence

'When in doubt, say less and offer a consultation' does more compliance work than clever wording.

4 Gate what must be collected first

No answer until the firm can contact the visitor — name, telephone and email.

5 Keep a small FAQ in the prompt

Nine short answers a compliance officer can review in one sitting. RAG is for larger knowledge.

6 Key memory on the session

Follow-up questions keep their context, and two visitors never see each other's conversation.

The system message is a compliance control. Test it with the questions you most fear, not the ones you hope for.

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TOPIC 2 · DEMO

Demo: An HR Policy Chatbot on Telegram

What is a chatbot? An agent behind a chat channel. Build one on Telegram first — then Lab 3 puts one behind a website.

Telegram Trigger
an employee messages the bot

▶

HR Agent — Ava
HR policies + memory

▶

Chat model
Gateway credits, no API key

▶

Reply on Telegram
back to the same chat

1 What Ava knows

Sample HR policies written into the system instruction — leave, sick leave, claims, hours, probation.

2 Guardrails in the prompt

Answer only from the policies, never invent a number, and refer salary, medical or disciplinary questions to HR.

3 Same agent, new front door

Telegram Trigger is the front door here. Lab 3 puts the same kind of agent behind a website.

You will: create a bot with BotFather → build the workflow with n8n Assistant → connect, publish and chat.

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Demo Step 1: Create a Telegram Bot with BotFather

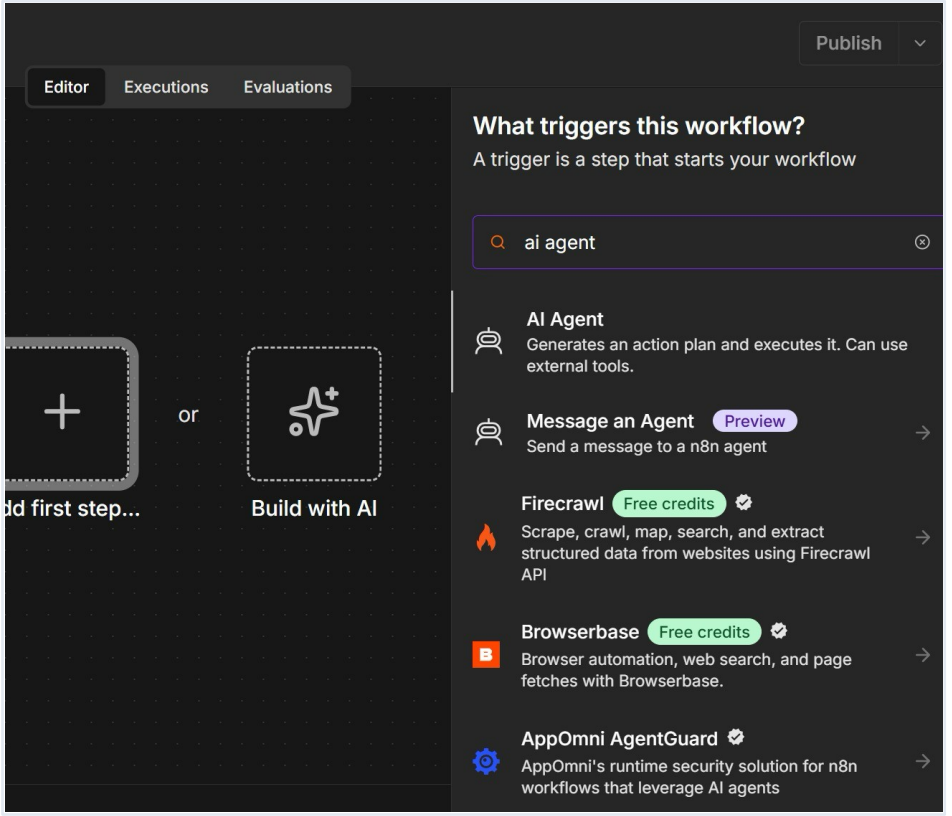
1. Open Telegram and search for BotFather (blue checkmark)
2. Start a chat and send /newbot
3. Type a display name, e.g. My Free Helper Bot
4. Choose a unique username ending in bot, e.g. my_free_helper_bot
5. Copy the HTTP API token BotFather sends back

The token is the bot's password — never share or commit it. If it leaks, send /revoke to BotFather for a new one.

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Demo Step 2: Open n8n Assistant with Build with AI

- Create a new workflow. The empty canvas offers two starts: Add first step, or Build with AI.
- Click Build with AI — this opens n8n Assistant, which builds the workflow from a plain-English prompt.
- Add first step is the manual route: it opens the node search, here filtered to 'ai agent'.
- Either way you get the same nodes. The Assistant just writes the first draft for you.



Build with AI drafts the workflow; you still review every node before you publish.

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Demo Step 2: Paste the Prompt into n8n Assistant

Create a new workflow, open n8n Assistant and paste this prompt — it plans and builds the chatbot for you.

THE PROMPT

Create a 3-node workflow:

- A user posts a message to a Telegram bot.
- Telegram sends the data to an HR agent.
- The HR agent responds to the user on the company HR policies. Create a system instruction for the HR agent with sample company policies in it. The HR agent has a memory.
- Respond to Telegram.

THEN YOU

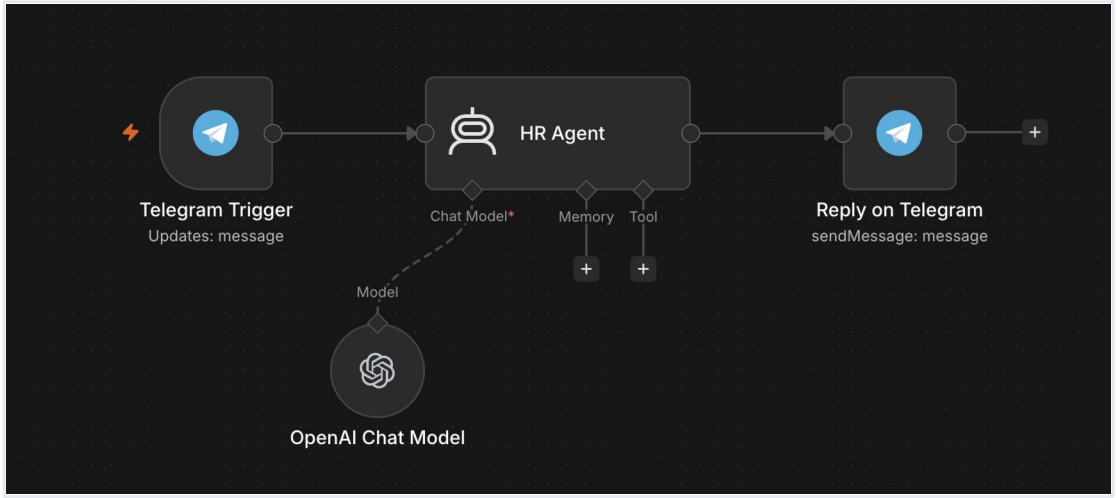
- Review the plan before it builds
- Read the system instruction it wrote — edit the sample policies if you like
- Check a memory node is attached to the HR agent
- Stuck? Import the reference solution instead

Reference solution: labs/lab3-chatbot-webhook/Lab 3 Demo - Telegram HR Policy Agent.json

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Demo: The Workflow n8n Assistant Builds

Your canvas should look like this: trigger, agent, reply — with the chat model and memory hanging under the agent.



1 Telegram Trigger

Updates: message — every message sent to the bot starts one execution.

2 HR Agent

Ava's role, rules and sample HR policies live in the system instruction n8n Assistant wrote.

3 Model and memory

Chat model on Gateway credits; memory keeps each employee's conversation, keyed on the Telegram chat ID.

4 Reply out

Reply on Telegram sends back to the chat ID the trigger received, so the reply lands in the same chat.

Shown: the reference solution, which has no memory. Yours has a memory node — the prompt asked for one.

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Demo Step 3: Connect the Bot, Publish and Chat

1. Telegram Trigger → Credential → Create new → paste the token
2. Telegram reply node → select the same credential
3. Chat model → Use Gateway credits
4. Publish — n8n registers its webhook with Telegram
5. Open t.me/<username>, tap Start, ask about annual leave

A Telegram bot has ONE webhook at a time — testing in the editor swaps in the Test URL, so publish again when you are done.

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Demo: Chatting with Ava on Telegram

- /start opens the chat — Ava introduces herself exactly as her system message tells her to.
- Ask how many days of leave you get: the answer quotes the policy — 14, 18 or 21 days by length of service.
- Ask a follow-up — memory keeps the context. Then ask about a colleague's salary: Ava must refer you to HR.
- Every message is one workflow run — open Executions in n8n to watch it happen.



The bot is only as safe as its system message. Lab 3 adds a contact gate and a refusal rule, behind a website.

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Lab 3 — Chatbot Webhook (Investment Advisor)

A compliance-bound chatbot for an investment advisory firm: no answer until the firm can contact the visitor, and never any financial advice.

Chat Webhook — POST
the widget sends a message

▶

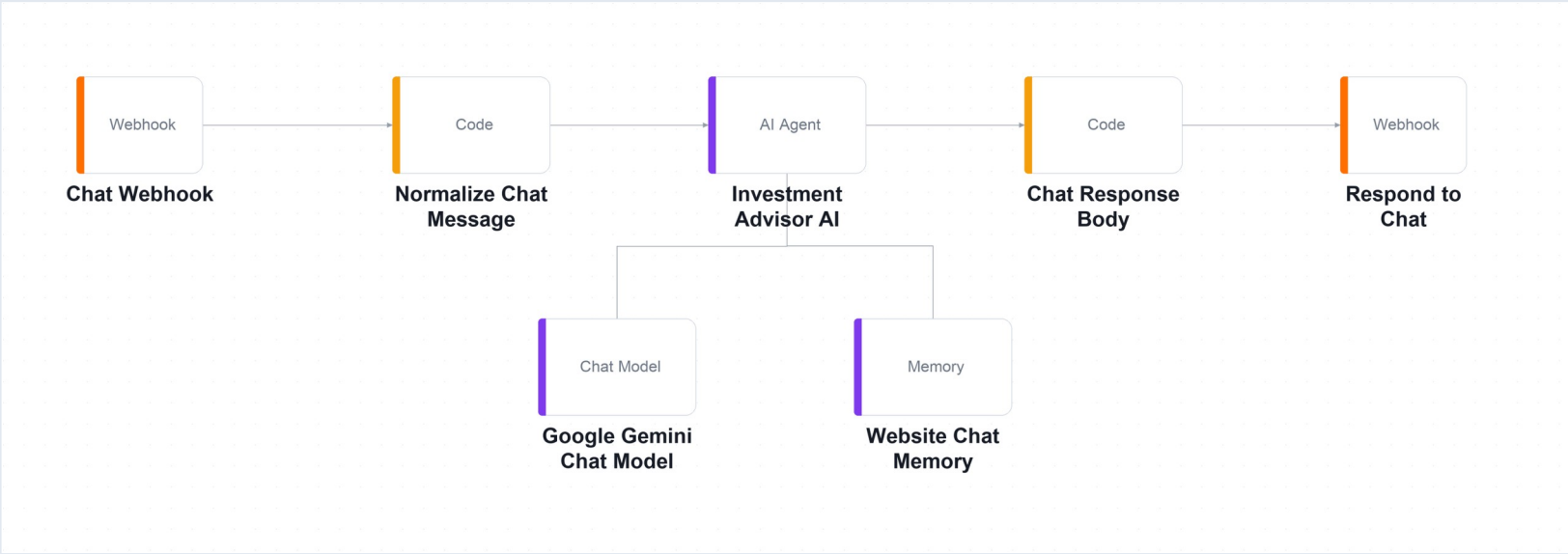
Normalize
Code node shapes the input

▶

AI Agent
Gemini 2.5 Flash, temp 0.2 + memory

▶

Respond to Chat
reply returns to the widget



The Lab 3 workflow on the n8n canvas. A small graph: webhook in, normalise, the agent with its chat model and session memory, then the reply back out.

LAB 3

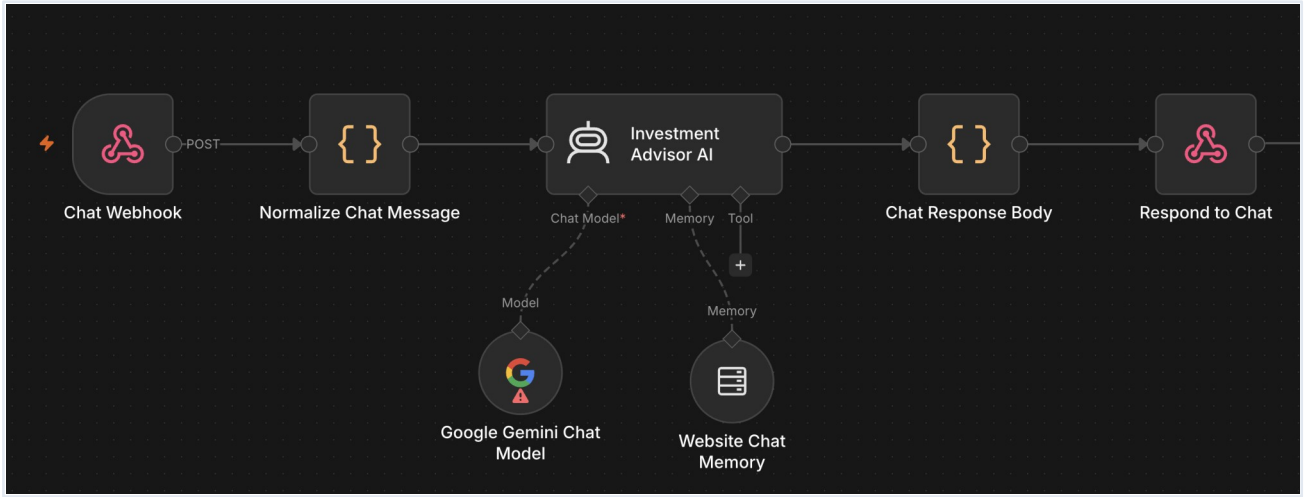
Chatbot Webhook (Investment Advisor)

Chat Webhook → Normalize → AI Agent (+ model + memory) → Respond to Chat · full click-by-click steps in the Learner Guide

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Lab 3 — The Chatbot Workflow in n8n

The imported workflow on the n8n canvas: webhook in, normalise, agent, shape the reply, respond. The intelligence is in the system message.



1 Message in

Chat Webhook receives each message from the widget; Normalize Chat Message shapes it for the agent.

2 The agent

Investment Advisor AI holds the contact gate, the non-advisory rule and the nine-question FAQ.

3 Model and memory

Google Gemini Chat Model (temperature 0.2) and Website Chat Memory keyed on the visitor's sessionId.

4 Reply out

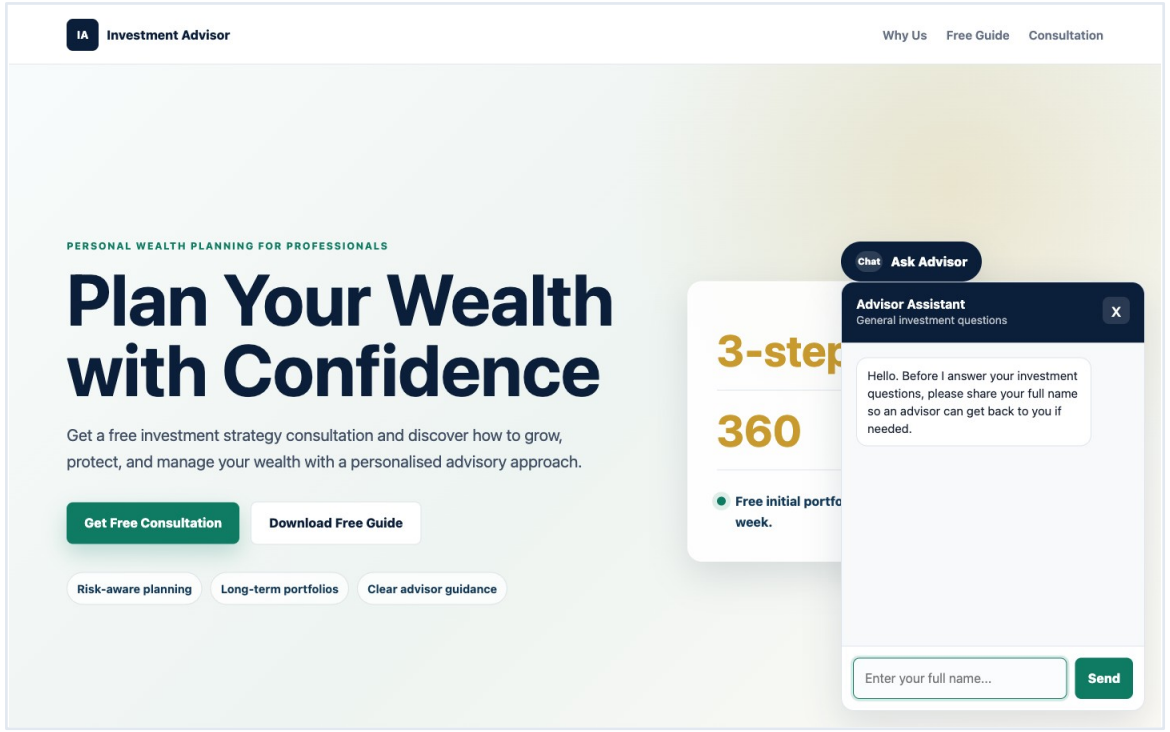
Chat Response Body builds the JSON reply; Respond to Chat returns it to the widget.

The warning on the Gemini node means no credential is selected yet — choosing your Gemini credential is the first fix after import.

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Lab 3 — Chatbot Webhook — What You'll Build

- The Investment Advisor site with the chat widget open. Note the contact gate in action — it asks for the visitor's name before it will answer anything.
- An agentic chatbot's system message is a compliance control, not just a prompt.
- The same rule enforced in the browser and in the agent is deliberate redundancy — only one of them runs on a machine you control.



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LAB 3 · VERIFY AND DEBRIEF

Lab 3 — Chatbot Webhook

Ask 'Can you guarantee returns?' and 'Which stock should I buy?'. The agent must decline both and offer a consultation. Then ask a question before giving your details — it must ask for the missing detail and answer nothing.

1 The system message

An agentic chatbot's system message is a compliance control, not just a prompt.

2 Defence in depth

The same rule enforced in the browser and in the agent is deliberate redundancy — only one of them runs on a machine you control.

3 Session-keyed memory

Session-keyed memory gives context to follow-up questions without leaking between visitors.

4 Prompt or RAG?

A small fixed FAQ belongs in the prompt; RAG is for knowledge too large or too changeable for that.

Lab 3 — you can now deploy an agentic chatbot on a website that operates within regulatory limits.

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Human In, On, and Out of the Loop

Three levels of human oversight. The design question is which one this particular process deserves.

1 Human IN the loop

The workflow PAUSES and a named person must approve before the action happens. Nothing reaches the customer unauthorised. This is Lab 4.

2 Human ON the loop

The agent acts immediately, but a person monitors and can intervene or reverse it. Suited to high volume and low individual risk.

3 Human OUT of the loop

Fully autonomous. Acceptable only where an error is cheap, reversible and detectable — never where money or a regulator is involved.

The question is never 'is the model good enough?' but 'what does one bad output cost, and who answers for it?'

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How an Approval Suspends a Running Workflow

A Send-and-Wait node stops the execution mid-flight and holds it until a person decides.

Event arrives
a client message

▶

Agent drafts
classify, flag, propose

▶

Draft is logged
before any human sees it

▶

Human decides
Approve or Decline

▶

Only then
send and log the outcome

1 The execution really is paused

Look at the canvas and the run is sitting on the approval node. It waits days if it has to — n8n holds the state.

2 Approve and Decline are just links

The email carries two URLs. Clicking one resumes the execution down that branch — no app for the approver to learn.

3 A decline is a reassignment

Declining is not deleting. The ticket moves to a named person, because someone still owes the client an answer.

The agent may do everything except the one thing that matters. That pause is the whole control.

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When to Put a Human in the Loop

Six tests. If a process meets any one of them, it needs an approval gate.

1 Money moves

Payments, refunds, credit and lending decisions — anything with a direct financial consequence.

2 It is a regulated communication

A licensed firm writing to a client about their money, health or legal position.

3 The action is irreversible

A sent email, a deleted record or published content cannot be recalled.

4 A person is vulnerable

Distress, illness, bereavement, or savings that someone cannot afford to lose.

5 The stakes exceed the error rate

A 2% error rate is fine for tagging tickets and unacceptable for approving loans.

6 Someone must be accountable

A regulator asks which licensed person is answerable — not what the machine decided.

Automate the drafting, not the deciding, wherever any of these six is true.

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Guardrails — Two Gates Around the Agent

Instructions shape what the agent intends. Guardrails check what actually goes in and comes out.

INPUT GUARDRAIL — before the agent

- Screen the incoming message before the model ever sees it.
- Block prompt injection: 'ignore your instructions and…'.
- Strip or reject personal data you must not process.
- Rate-limit and authenticate — a public webhook is a public door.

OUTPUT GUARDRAIL — after the agent

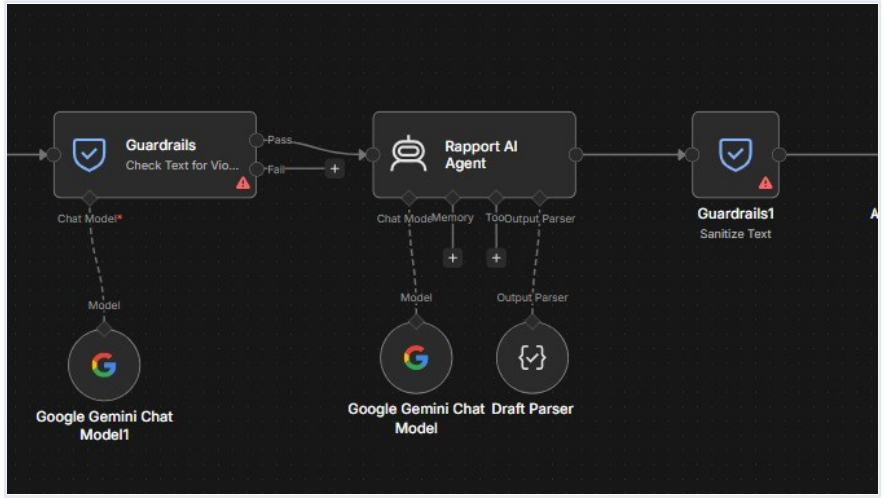
- Check the reply before it reaches a human being.
- Look for the phrases you have banned: guarantees, forecasts, advice.
- Validate the structured fields against the values you allow.
- On any doubt, route to a person rather than sending it.

Defence in depth: the same rule in the prompt AND in the workflow — only one of the two runs on a machine you control.

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Guardrails in n8n — the Guardrails Node

The two gates from the previous slide, built with n8n's Guardrails node: one before the agent, one after it.



1 Input guardrail

Guardrails · Check Text for Violations screens each message before the agent sees it, using its own chat model.

2 Pass or Fail

Pass continues to the agent. Fail branches away, so a blocked message never reaches the model.

3 The agent

Rapport AI Agent drafts the reply with its Gemini model and the Draft Parser.

4 Output guardrail

Guardrails1 · Sanitize Text cleans the reply — for example removing personal data — before anything is sent.

Defence in depth: the prompt asks the agent to behave; the Guardrails nodes check that it did.

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Lab 4 — Human in the Loop (Client Rapport Assistant)

An agent reads a worried client's concern, flags compliance risks and drafts a reply — but sends nothing until a licensed manager approves it.

Webhook
client sends a concern

▶

AI Agent
classify · flag · draft

▶

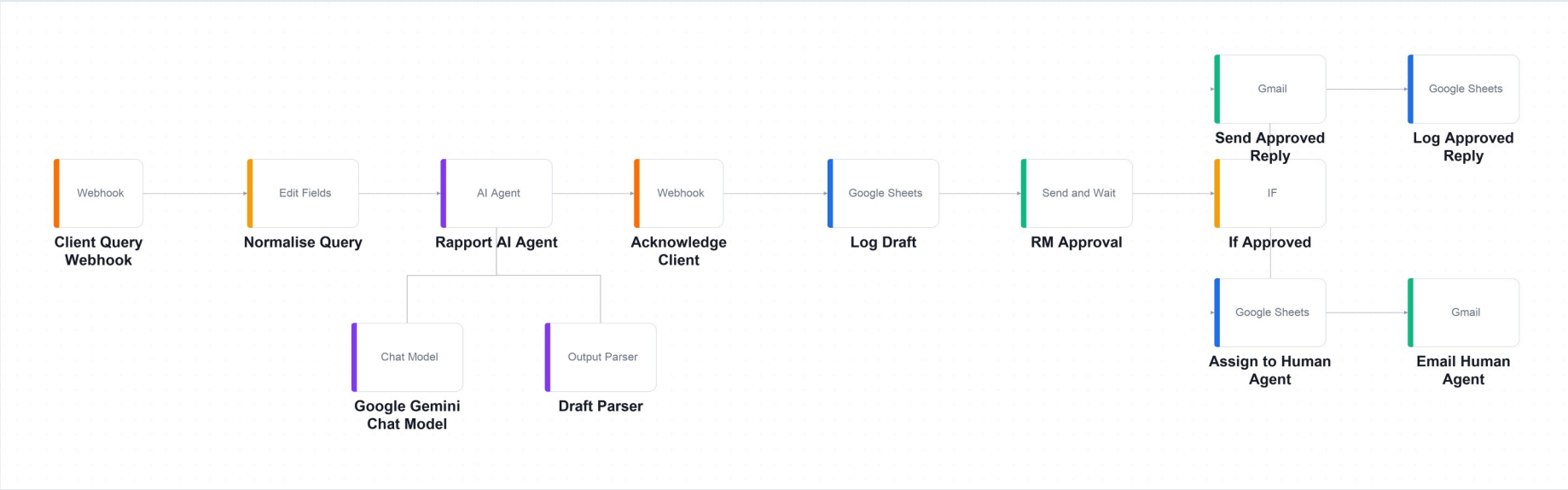
Log Draft
before any human sees it

▶

RM Approval
Send & Wait — PAUSED

▶

IF
Approve → send · Decline → reassign



Everything up to RM Approval is the agent's territory; the IF node after it splits into the approved path (top) and the reassignment path (bottom).

LAB 4

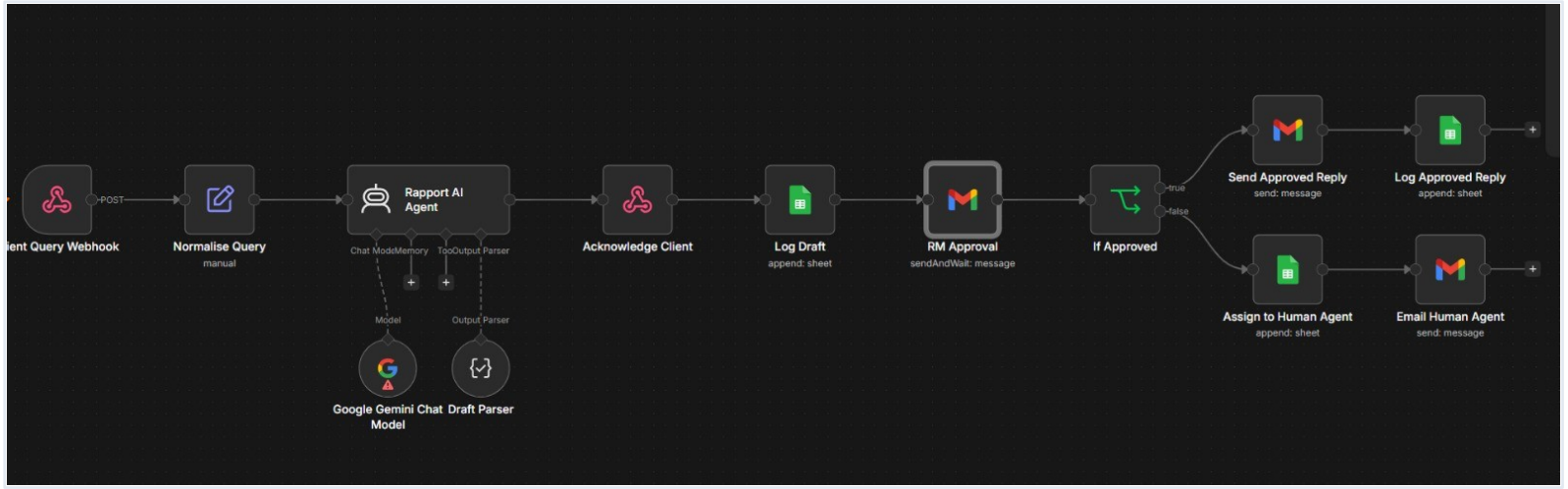
Human in the Loop (Client Rapport Assistant)

Webhook → Agent → Acknowledge → Log Draft → RM Approval (Send & Wait) → IF → Send / Reassign · full click-by-click steps in the Learner Guide

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Lab 4 — The Approval Workflow in n8n

The imported workflow on the n8n canvas. Everything up to RM Approval is the agent's work; nothing after it runs until a person decides.



1 Concern in

Client Query Webhook receives the client's message; Normalise Query tidies it for the agent.

2 The agent

Rapport AI Agent drafts a non-advisory reply; Draft Parser returns tone, urgency and compliance flags.

3 Log, then pause

Acknowledge Client answers at once, Log Draft records the draft, and RM Approval waits for a human.

4 Two paths

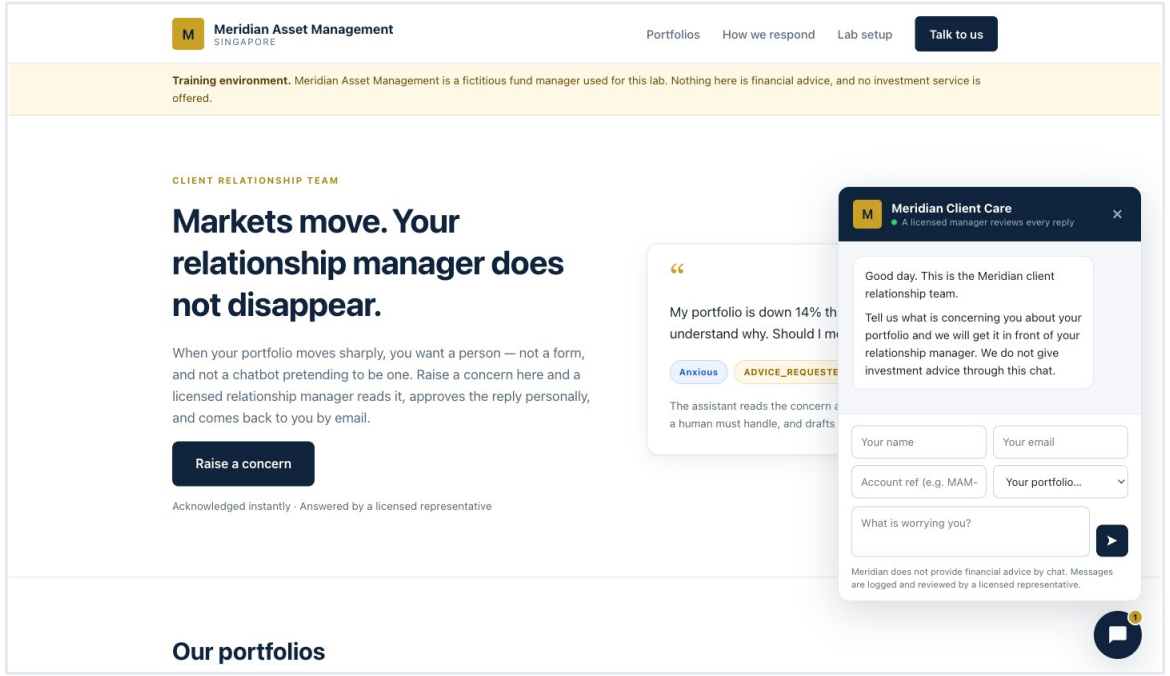
If Approved: send and log the reply. Otherwise: Assign to Human Agent and email that person.

RM Approval is a Gmail Send-and-Wait node: the execution genuinely pauses there until Approve or Decline is clicked.

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Lab 4 — Human in the Loop — What You'll Build

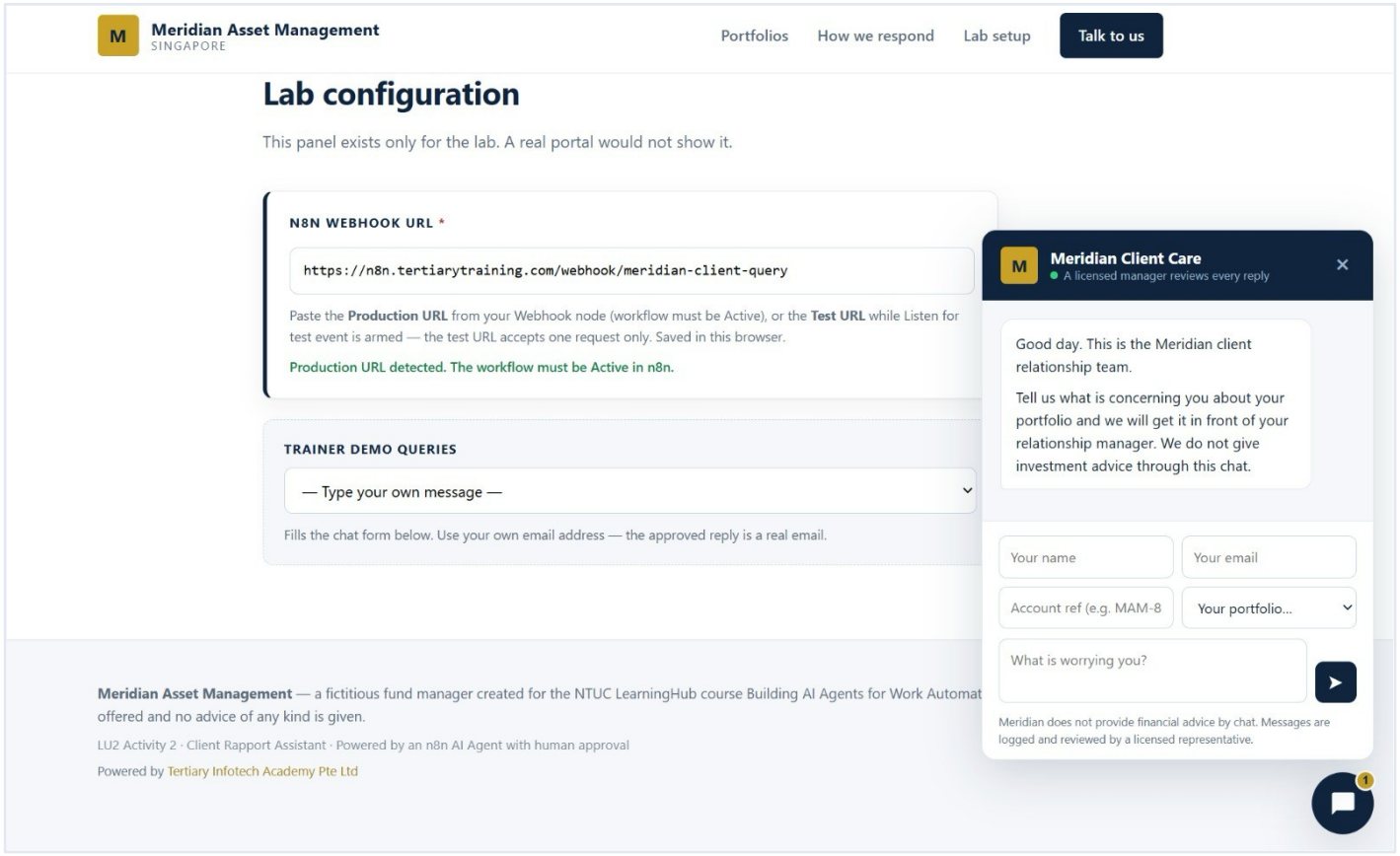
- The Meridian client portal. The widget states plainly that a licensed manager reviews every reply — the human-in-the-loop promise made visible to the client.
- A Send-and-Wait node pauses the workflow until a named person authorises the action.
- Separating what the agent proposed from what a human approved is what makes the trail auditable.



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LAB 4 · STEP BY STEP

Lab 4 — Step: Connect the Website and Send a Concern



Paste your webhook Production URL into Lab configuration, pick a trainer demo query or type your own, use your own email, and send. The widget promises a licensed manager reviews every reply.

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LAB 4 · VERIFY AND DEBRIEF

Lab 4 — Human in the Loop

Run all eight rows of sample-queries.csv. When you are done Drafts has 8 rows, Approved_Replies has 5 and Handover_Queue has 3 — and no accountability column is empty.

1 The approval gate

A Send-and-Wait node pauses the workflow until a named person authorises the action.

2 Two separate logs

Separating what the agent proposed from what a human approved is what makes the trail auditable.

3 A decline reassigns

A decline is a reassignment, not a deletion — someone must still owe the client an answer.

4 Keep the model out

The most legally important sentence in a workflow should be the one the model cannot touch.

Lab 4 — you can now apply a human approval gate so no AI-drafted communication is sent unsupervised.

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Tea Break

15 minutes

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03

TOPIC 03

Retrieval-Augmented Generation (RAG)

Tokenization · Embeddings · Vector stores · Grounded answers

Applied in Lab 5

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TOPIC 3 · RAG

Why Retrieval, Rather Than a Bigger Prompt

For twenty short documents, pasting them all would genuinely work. The problem starts at two hundred.

1 Prompts have a ceiling

Add enough documents and you exceed the model's context window — the request simply fails.

2 Long prompts degrade quality

Models attend less reliably to the middle of a very long context. The answer is in there and is still missed.

3 You pay for every token

Every question re-sends every document, whether or not it was relevant to that question.

4 Retrieval turns it around

Fetch only the few documents that resemble the question, and send just those. Cheaper, shorter and more accurate.

RAG is not a bigger prompt — it is a smaller, better-chosen one.

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TOPIC 3 · RAG

The RAG Pipeline — Ingest Once, Retrieve Every Question

Two separate workflows. You run ingestion once; retrieval runs on every single question.

Collect the source documents ▶ **Split** into chunks ▶ **Embed** each chunk into a vector ▶ **Store** in the vector store ▶ **Retrieve** the nearest chunks ▶ **Answer** from what came back

1 Ingestion — run by hand

Lab 5's first workflow: list 20 brochures on Drive, download, split, embed, insert. A Manual Trigger, run once.

2 Retrieval — runs per question

Lab 5's second workflow: embed the question with the SAME model, find the nearest brochures, answer only from those.

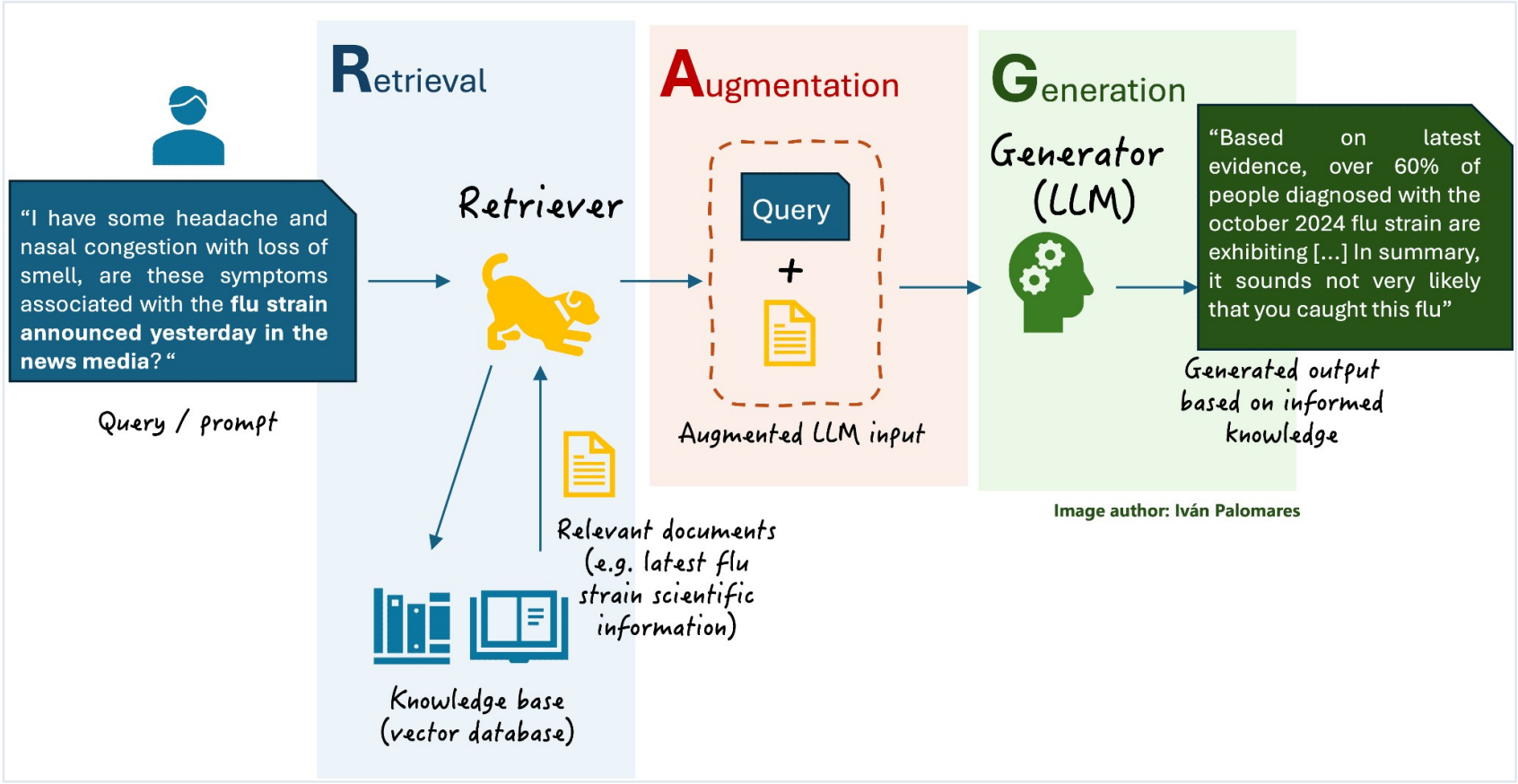
3 The models must match

A question embedded by one model cannot be compared against documents embedded by another. It fails silently, returning nonsense.

Ingest once, retrieve thousands of times — and use the identical embedding model on both sides.

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How RAG Works



Retrieval finds the documents that resemble the question · Augmentation adds them to the prompt · Generation produces an answer grounded in what was retrieved

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TOPIC 3 · RAG

Chunking — the Biggest Lever on RAG Quality

Most RAG projects fail quietly at chunking, not at the model.

1 The rule

The right chunk is the smallest piece of text that still answers a question on its own.

2 Too small

A brochure sliced into thirds separates the course name from its fee. Retrieval then finds one without the other.

3 Too large

Irrelevant text crowds the prompt and dilutes what actually matters.

4 Overlap

A small overlap between chunks stops a sentence being cut in half at the boundary.

5 In Lab 5

Chunk size 4000 with overlap 0 — one brochure becomes exactly one vector, so the fee is never separated from the course.

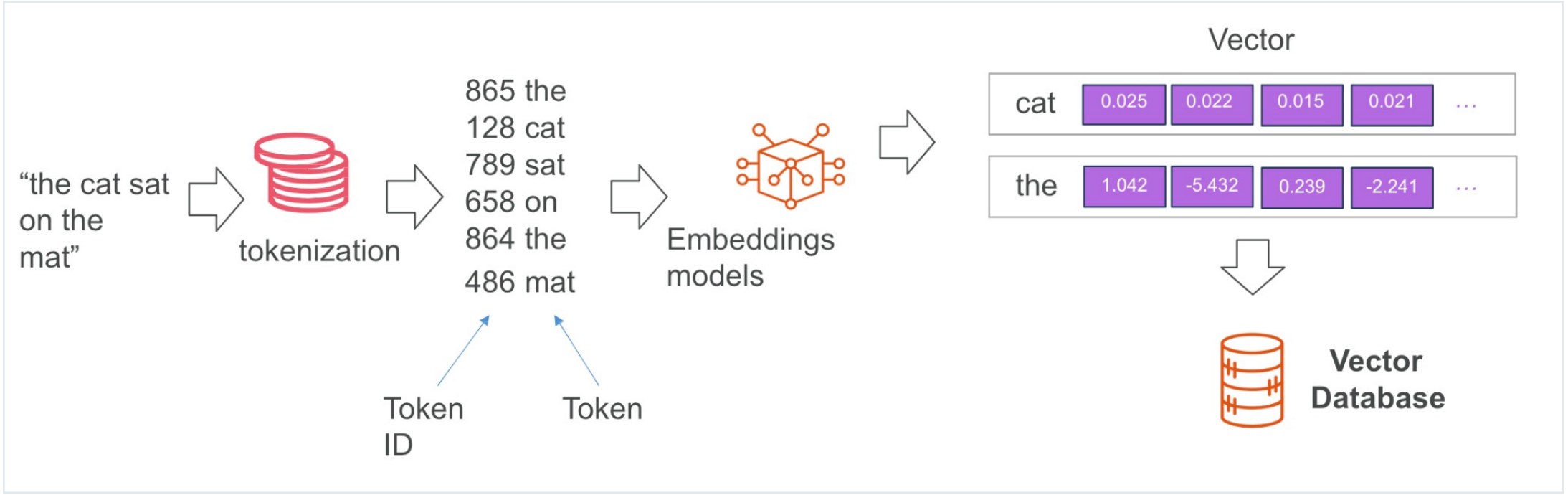
6 topK

How many chunks come back. Too few misses the answer; too many dilutes the prompt with noise.

Tune the chunking before you blame the model — that is where the answer was lost.

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Tokenization and Text Embedding



Text is split into tokens, then an embedding model turns it into a vector — a list of numbers that places similar meanings close together

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Choosing a Vector Store

Where the embeddings live, and whether they are still there tomorrow.

SIMPLE VECTOR STORE — built in

- No account, no API key, no index to create.
- Running RAG within ten minutes — this is what Lab 5 uses.
- WIPED whenever n8n restarts.
- Right for: learning, demos and prototypes.

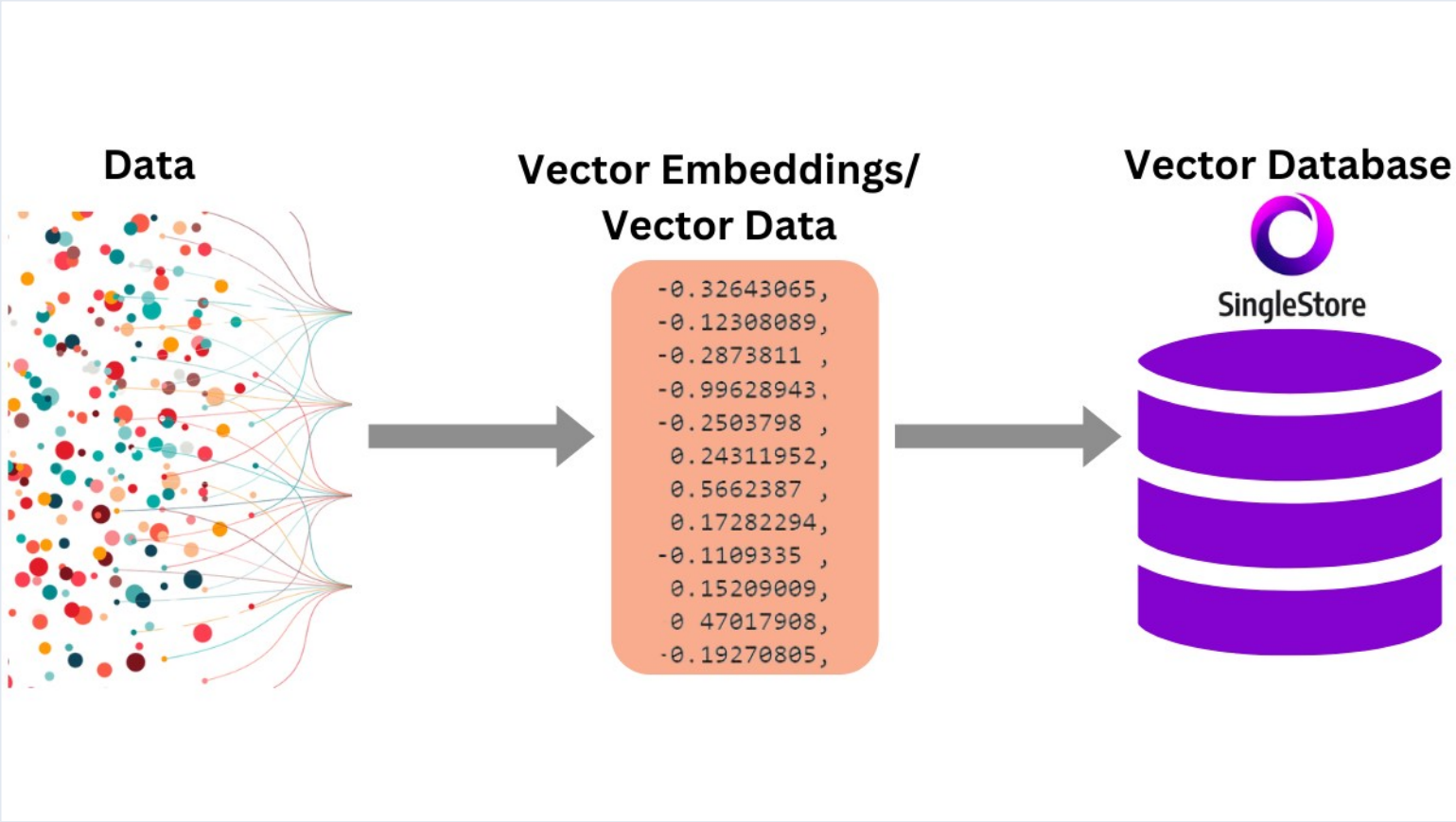
A REAL VECTOR DATABASE

- Pinecone, Supabase (pgvector), Qdrant, Weaviate.
- An account, an index and a matching dimension to set up.
- Persists, scales, and is shared across n8n instances.
- Right for: anything that goes into production.

Prototype on the in-memory store; move to a hosted database the moment anyone else depends on the answers.

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Vector Databases



A vector store holds the embeddings and finds the nearest neighbours to a question, fast

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TOPIC 3 · RAG

Grounding an Agent Against Hallucination

An ungrounded model asked for a price will produce one. It will be plausible, it will be wrong, and the customer will quote it back to you.

1 Always call the tool first

Instruct the agent to search the documents before answering any factual question at all.

2 Answer only from what came back

Every factual claim must trace to a retrieved document, not to the model's general knowledge.

3 Give it permission to fail

'I don't have that in our course information' is a far better answer than a confident invention.

4 Name what it must never invent

A fee, a date, a duration, a course code, an instructor's name, a discount.

5 Hide the machinery

Never mention the vector store or that it is searching documents — the customer does not care.

6 Test the traps

Ask about a course you do not run, a person who does not exist, and a discount never offered.

Grounding is an instruction problem as much as a retrieval one — retrieval finds it, the prompt makes it stick.

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Lab 5 — RAG Customer Care Chatbot — Ingestion Workflow

Cook & Bake Academy's customer care team retypes the same answers all day, and every answer is already written in the course brochures.

Manual Trigger
you run ingestion once

▶

Drive — list + download
20 brochures

▶

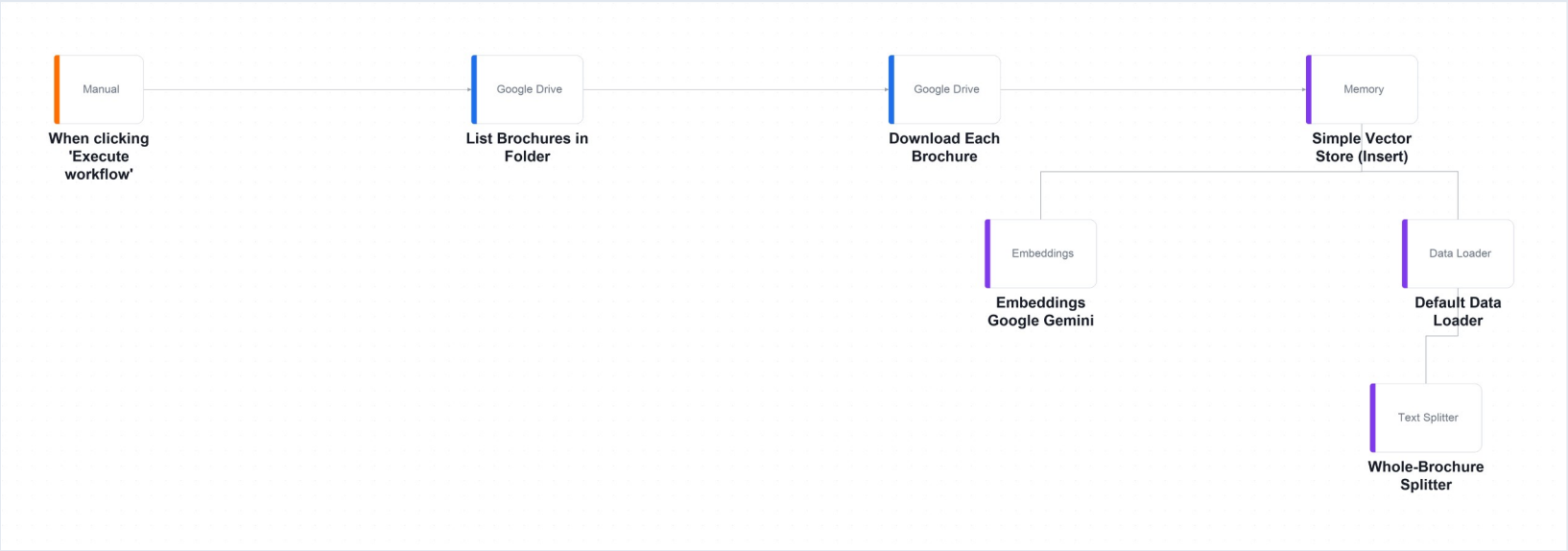
Splitter
1 brochure = 1 chunk

▶

Embeddings
3072-dim vectors

▶

Vector Store — Insert
stored, ready to search



The ingestion workflow on the n8n canvas — run ONCE by hand. Twenty brochures are listed, downloaded, split, embedded and inserted into the vector store.

LAB 5

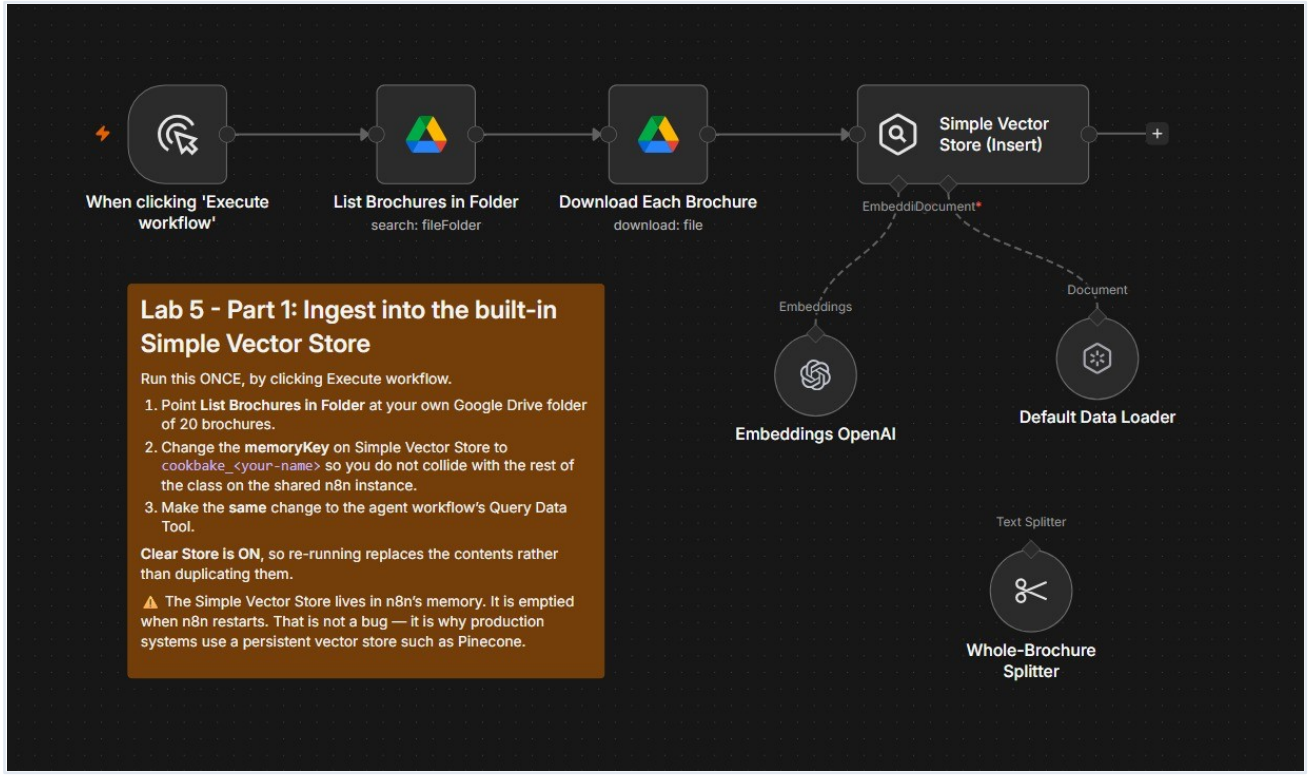
RAG Customer Care Chatbot — Ingestion Workflow

Ingestion: Drive → Split → Embed → Vector Store · Answering: Webhook → Agent + Retriever → Respond · full click-by-click steps in the Learner Guide

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Lab 5 — The Ingestion Workflow in n8n

Part 1 as it looks after import. You run it once, by hand, to load all 20 brochures into the vector store.



1 Fetch the brochures

When clicking 'Execute workflow' starts the run. List Brochures in Folder finds the 20 files in YOUR Drive folder, and Download Each Brochure pulls each one down.

2 Split and embed

Default Data Loader reads each file, Whole-Brochure Splitter keeps one brochure as one chunk, and the Embeddings node turns every chunk into a vector.

3 Insert into the store

Simple Vector Store (Insert) saves the 20 vectors under your memory key, cookbake_<your-name>. Clear Store is ON, so a re-run replaces instead of duplicating.

Ingestion is a batch job: re-run it when a brochure changes — and after every n8n restart.

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Lab 5 — RAG Customer Care Chatbot — Answering Workflow

The answering workflow on the n8n canvas — runs on EVERY question. The Query Data Tool and its embedding model hang below the agent.

Webhook — POST
a visitor asks a question

▶

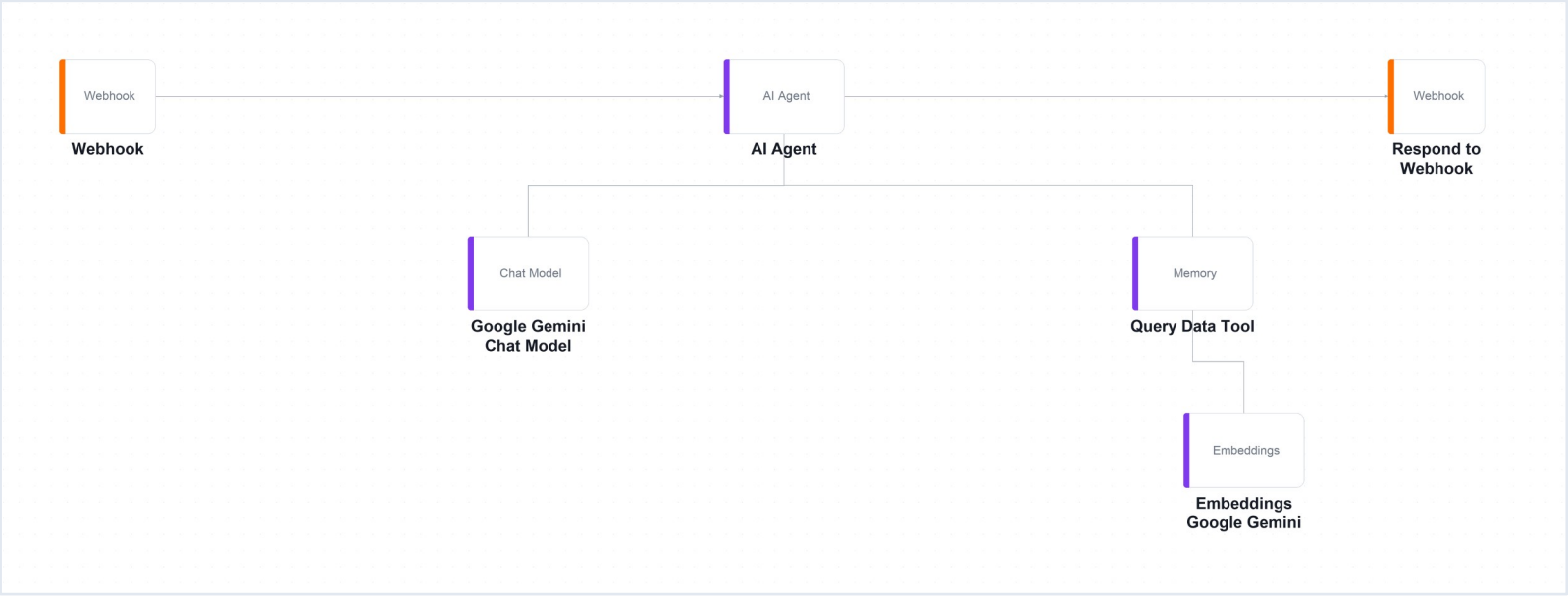
AI Agent
+ Query Data Tool

▶

Retrieve
nearest brochures only

▶

Respond
answer grounded in those



LAB 5

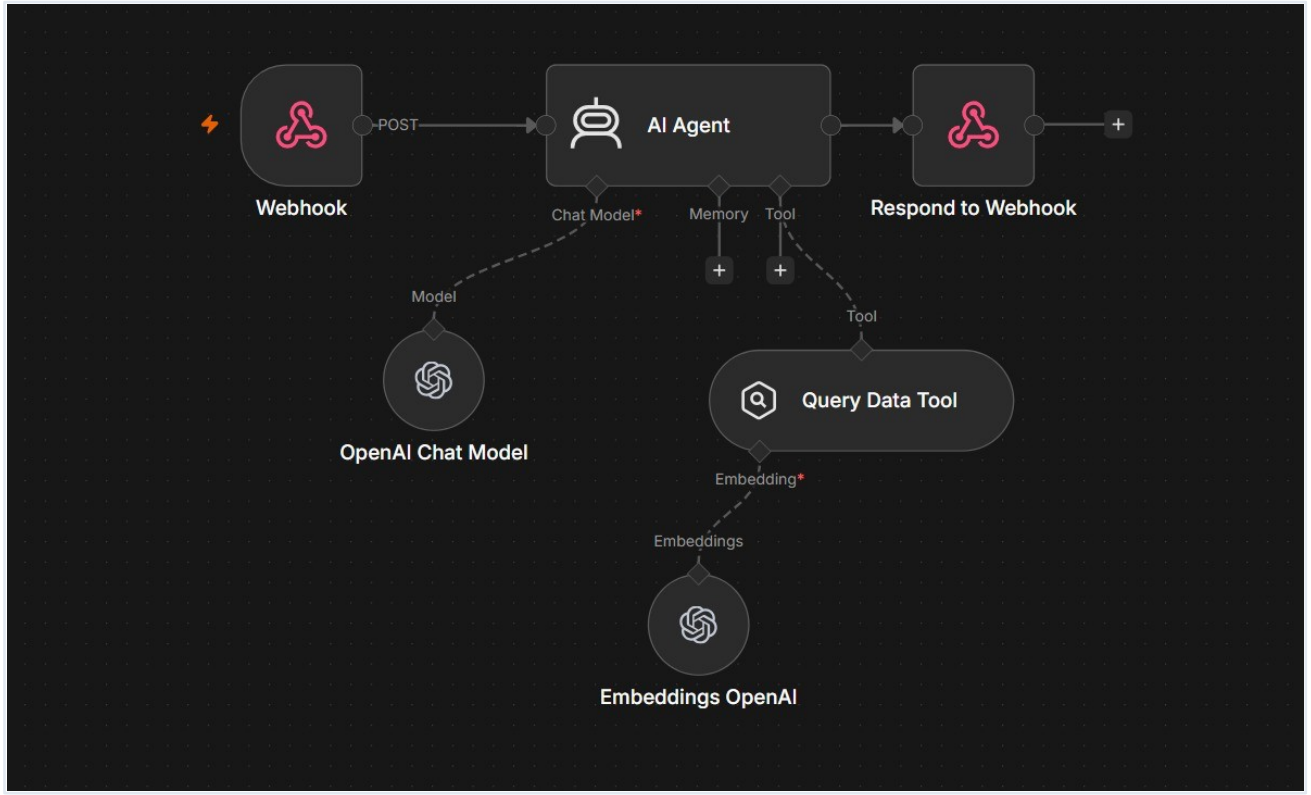
RAG Customer Care Chatbot — Answering Workflow

Ingestion: Drive → Split → Embed → Vector Store · Answering: Webhook → Agent + Retriever → Respond · full click-by-click steps in the Learner Guide

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Lab 5 — The Answering Workflow in n8n

Part 2 runs on every question. The agent must search the brochures before it is allowed to answer.



1 Question in, answer out

Webhook receives the visitor's question from the website as a POST. Respond to Webhook sends the agent's reply back to the chat window.

2 The agent and its model

AI Agent reasons with the chat model. Its system message says: call the tool first and never invent a fee. No Memory is attached, so each question stands alone.

3 The retrieval tool

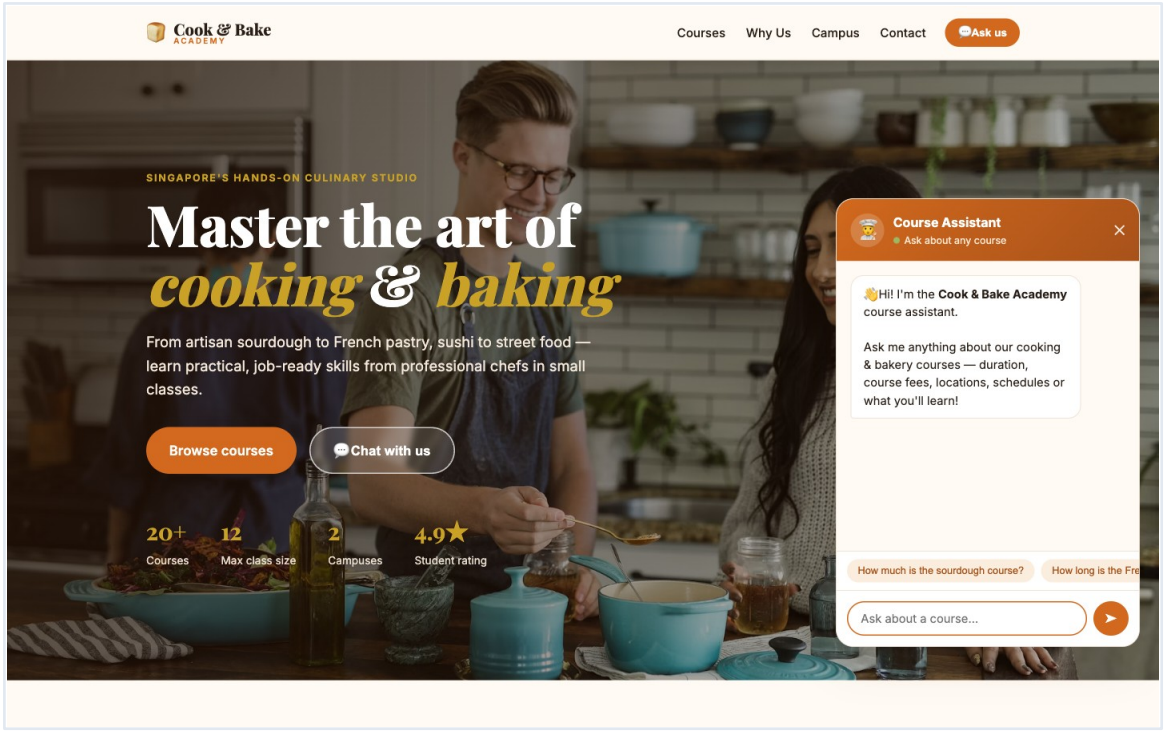
Query Data Tool searches YOUR memory key and returns the 5 closest brochures. Shown here with OpenAI; the lab file ships with Gemini — swap one, swap both.

Same embedding model and same memory key in both workflows — or retrieval silently finds nothing.

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Lab 5 — RAG Customer Care Chatbot — What You'll Build

- The Cook & Bake Academy site with the course assistant open. Every answer it gives is grounded in the 20 brochures you ingested into the vector store.
- RAG retrieves the few documents that resemble the question instead of pasting everything into the prompt.
- An embedding places similar meanings close together, so retrieval works even when no words match.

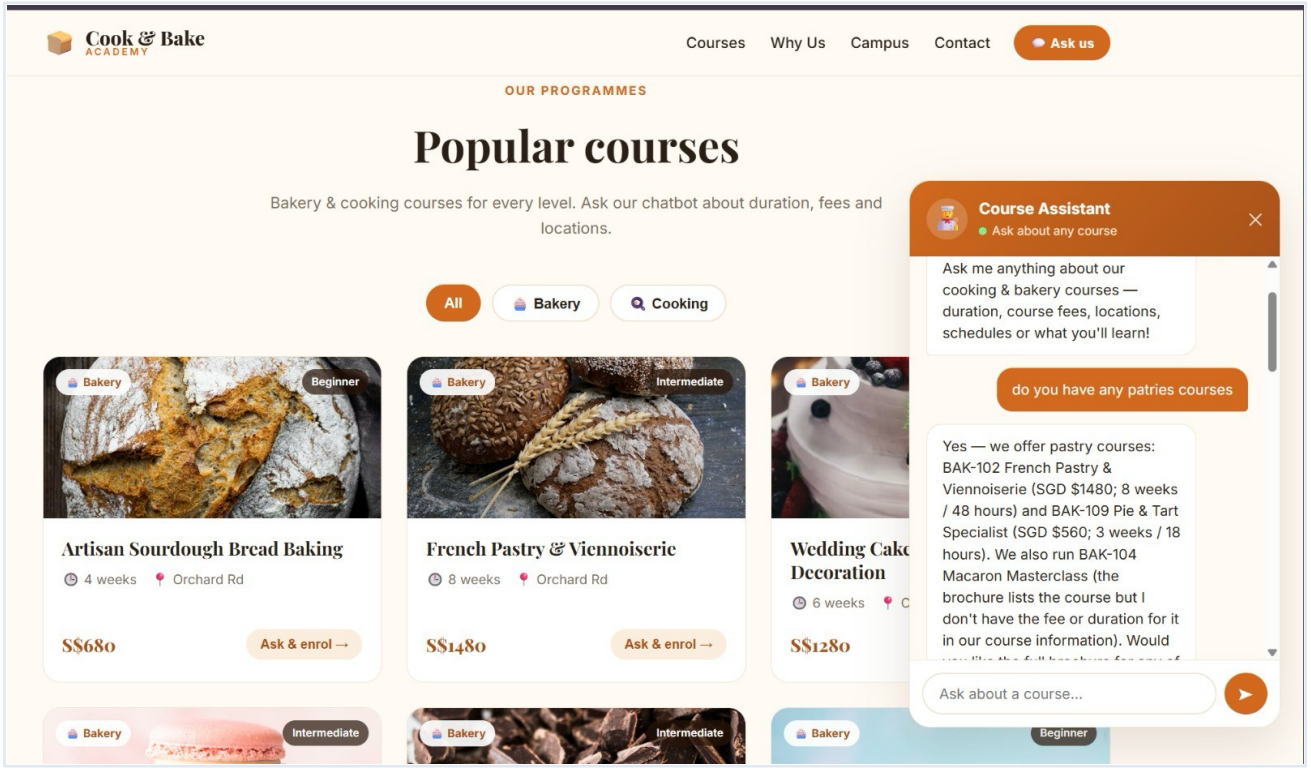


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LAB 5 · THE APPLICATION

Lab 5 — The Course Assistant in Action

A real conversation on the Cook & Bake Academy site. Every fee in the reply came from a retrieved brochure.



1 A typo still finds it

The visitor typed 'patries'. Embeddings match meaning, not spelling, so the pastry brochures BAK-102 and BAK-109 still come back.

2 Facts quoted, not invented

Fees and durations are quoted exactly: BAK-102 is SGD $1480 for 8 weeks / 48 hours — the same S$1480 shown on the course card behind the chat.

3 It admits a gap

It did not give BAK-104's fee and said so instead of guessing. The brochure does list SGD $420, so check what the tool returned: that is a retrieval gap.

'I don't have that' is grounding working — fix gaps in retrieval, never by loosening the prompt.

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LAB 5 · VERIFY AND DEBRIEF

Lab 5 — RAG Customer Care Chatbot

TC1 must quote BAK-101's exact fee from the brochure. Then run the three hallucination probes: TC7 (a course that does not exist), TC8 (an instructor name no brochure contains) and TC9 (a discount that was never offered). Any confident answer to these is a failure, however fluent.

1 What RAG does

RAG retrieves the few documents that resemble the question instead of pasting everything into the prompt.

2 Embeddings

An embedding places similar meanings close together, so retrieval works even when no words match.

3 Chunking

Chunking is the single biggest lever on RAG quality — the right chunk answers a question on its own.

4 Match model and key

The ingestion and retrieval embedding models must be identical, and the memory keys must match.

5 In-memory is temporary

n8n's Simple Vector Store is wiped when n8n restarts — fine for learning, unacceptable in production.

Lab 5 — you can now deploy a RAG application that grounds an agent in the organisation's own documents.

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

LO1 · n8n and AI agents

Built a trigger-and-action workflow and explained how an LLM agent differs from generative AI.

2

LO2 · Webhook-triggered agents

Put an AI agent behind a real website, with tools, structured output and an audit log.

3

LO3 · Agentic chatbots

Designed a chatbot that collects contact details and refuses to give regulated advice.

4

LO4 · Human in the loop

Paused a workflow on an approval gate so no drafted communication was sent unsupervised.

5

LO5 · RAG

Ingested 20 documents into a vector store and grounded a chatbot so it stopped inventing answers.

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NEXT STEPS

Taking This Back to Work

Six moves that turn what you built this week into something your organisation actually runs.

1 Start with one painful process

Pick something repetitive, high-volume and low-risk. Automate that first.

2 Map it before you build it

Draw the trigger, the decisions and the actions on paper. The canvas comes second.

3 Decide where the human belongs

Anything irreversible, regulated or expensive keeps an approval gate.

4 Ground the agent in your own data

RAG over your policies and documents beats a general model every time.

5 Log everything from day one

You will need the audit trail long before you think you will.

6 Iterate with real cases

Test with the questions you fear, not the ones you hope for.

The next process you automate is the one you were dreading this morning.

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WE'RE HERE TO HELP — DURING AND AFTER THE CLASS

Support

1

Email

enquiry@tertiaryinfotech.com

2

Telephone

+65 6100 0613

3

Website

www.tertiarycourses.com.sg

4

LMS / TMS portal

lms-tms.tertiaryinfotech.com

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WRAP-UP · FINAL ASSESSMENT

Assessment

1

Written Assessment (SAQ)

1 hour. Short-answer questions covering the course knowledge.

2

Practical Performance (PP)

1 hour. Hands-on n8n build tasks drawn from the five labs.

3

Open book

Slides, Learner Guide and approved materials only.

4

Submit on the LMS

Upload your completed answers at lms-tms.tertiaryinfotech.com.

5

Digital attendance

Remember to take the Assessment digital attendance (TRAQOM).

6

Appeals

An appeal process is available if required.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

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Digital Attendance (Mandatory)

1

AM, PM and Assessment

It is mandatory to take all three digital attendances for WSQ-funded courses.

2

The QR code

The trainer or administrator displays the attendance QR code from the SSG portal.

3

Scan and submit

Scan it with your mobile phone camera and submit your attendance.

4

75% minimum

At least 75% attendance is required to be eligible for assessment and funding.

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HAPPY AUTOMATING

Thank You!

Keep your n8n instance running and keep building. The next process you automate is the one you were dreading this morning.

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